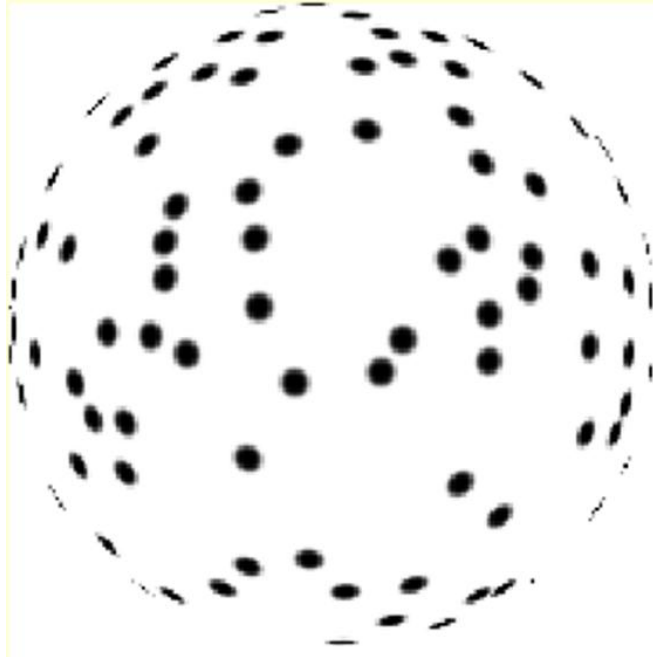
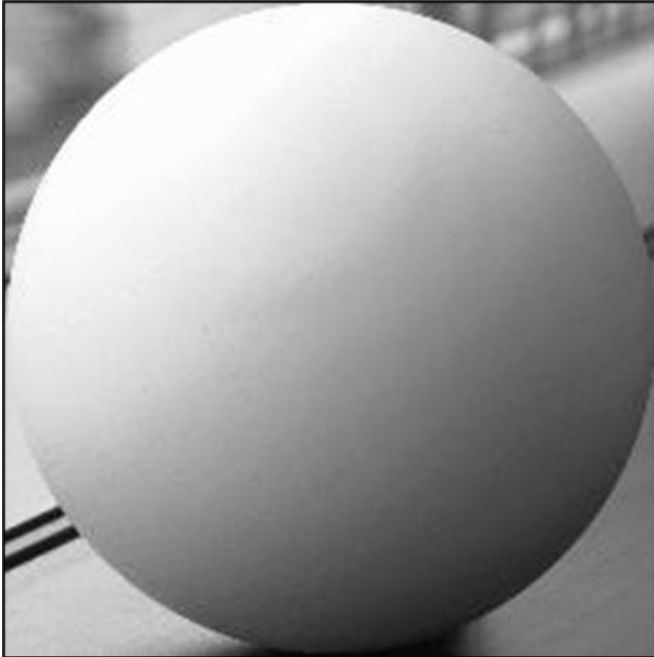


More Video Sources: Shree Nayar, Columbia U

- <https://www.youtube.com/@firstprinciplesofcomputerv3258>
- **First Principles of Computer Vision**
- Reflectance Models; Shape from Shading; Camera Calibration
- Video clips covers the entire course!

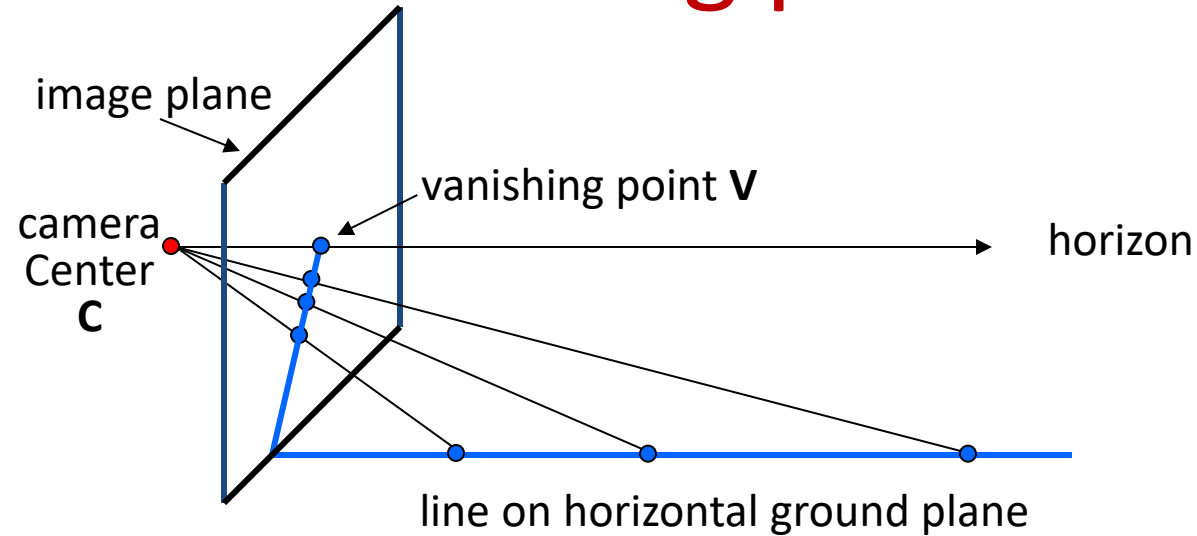
3D from One Camera (Monocular Cues)

- Last week we saw how to compute depth from 2 camera using stereo methods (Binocular Cues)
- 3D cues are also available in a single image from one camera.



Perspective Projection of Parallel lines

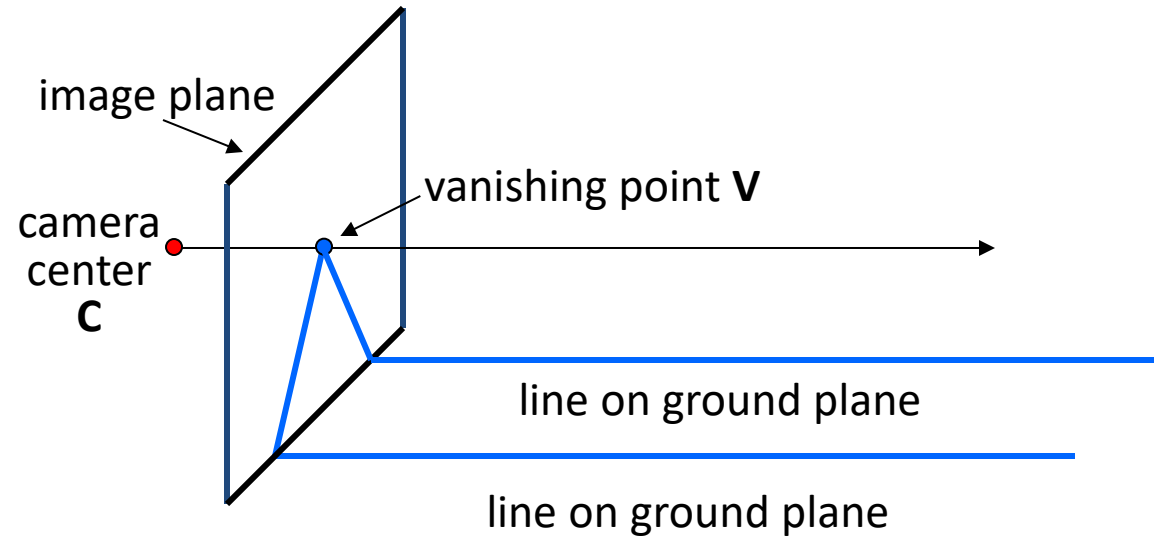
Vanishing points



Vanishing point & Horizon

- Horizon is the projection of a point at infinity
- Horizontal camera: horizon is at center of image
- Horizontal camera: horizon intersects objects at camera height

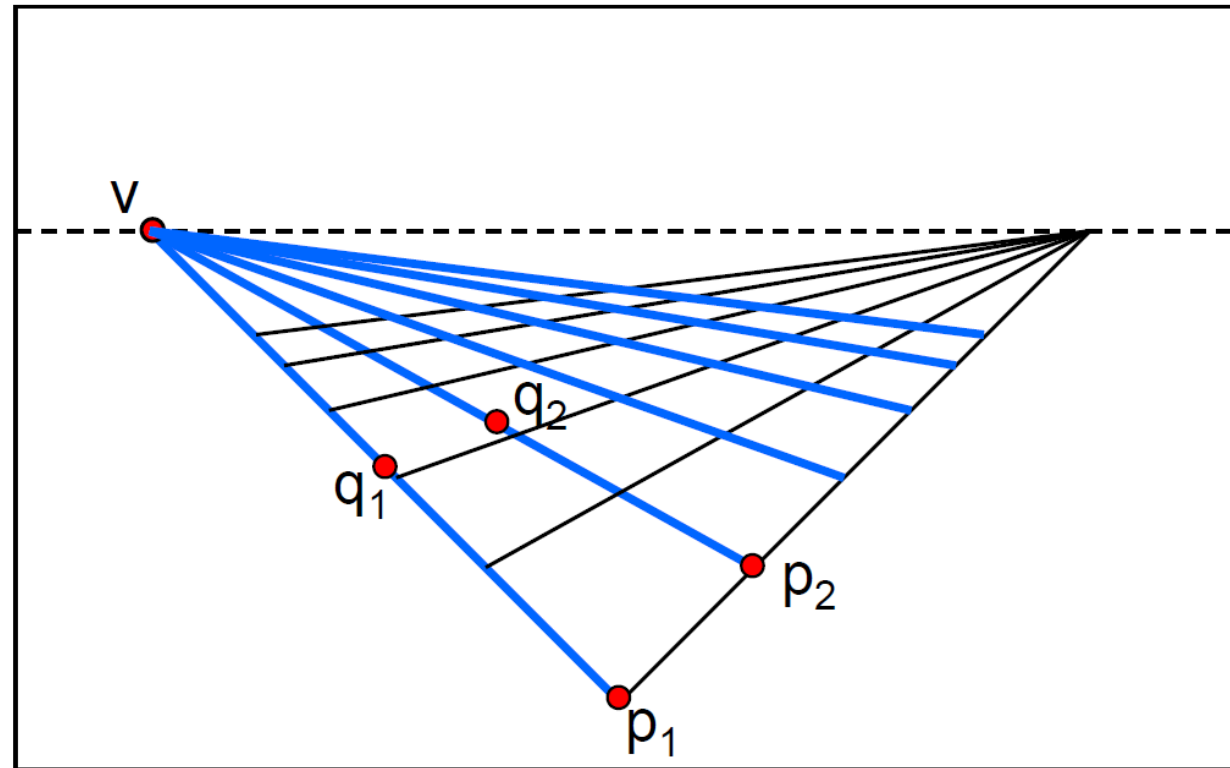
Vanishing points



Properties

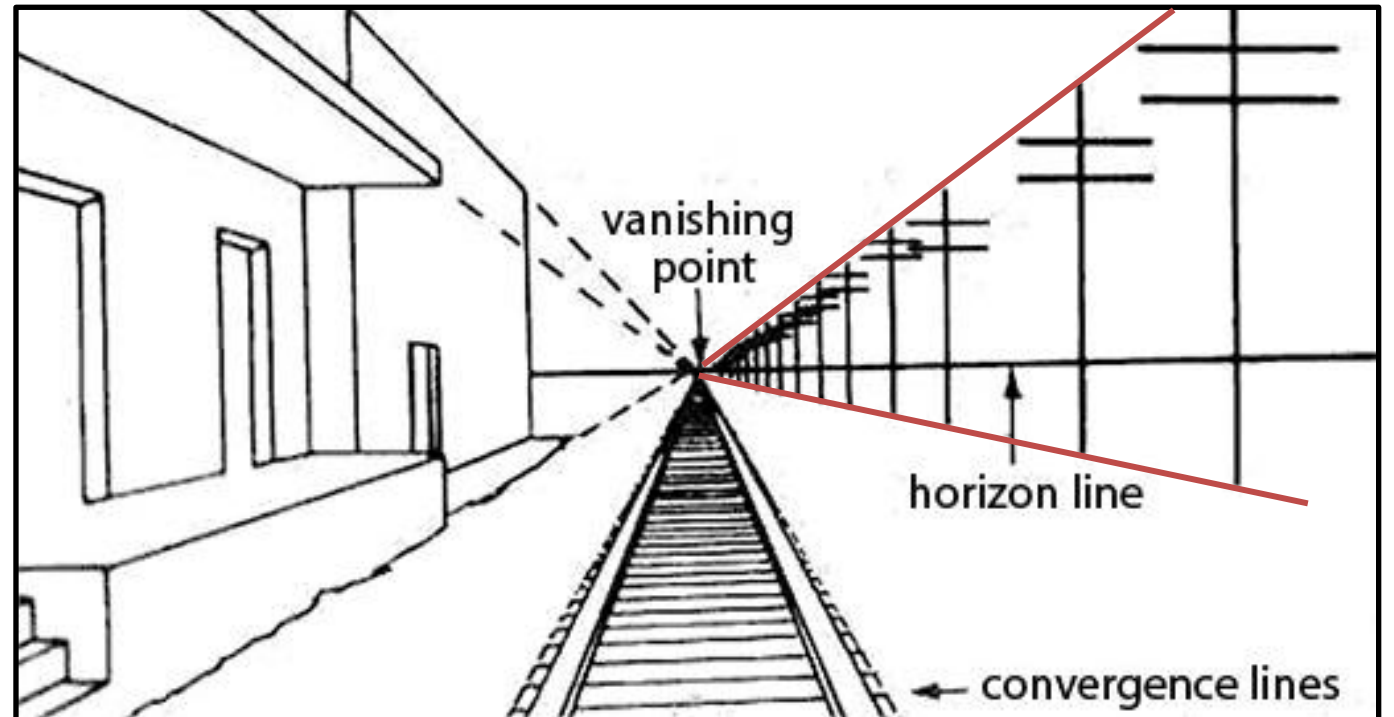
- Any two parallel lines have the same vanishing point v
- The ray from Camera Center C through v is parallel to these lines
- An image may have more than one vanishing point

Perspective Projection – Vanishing Points

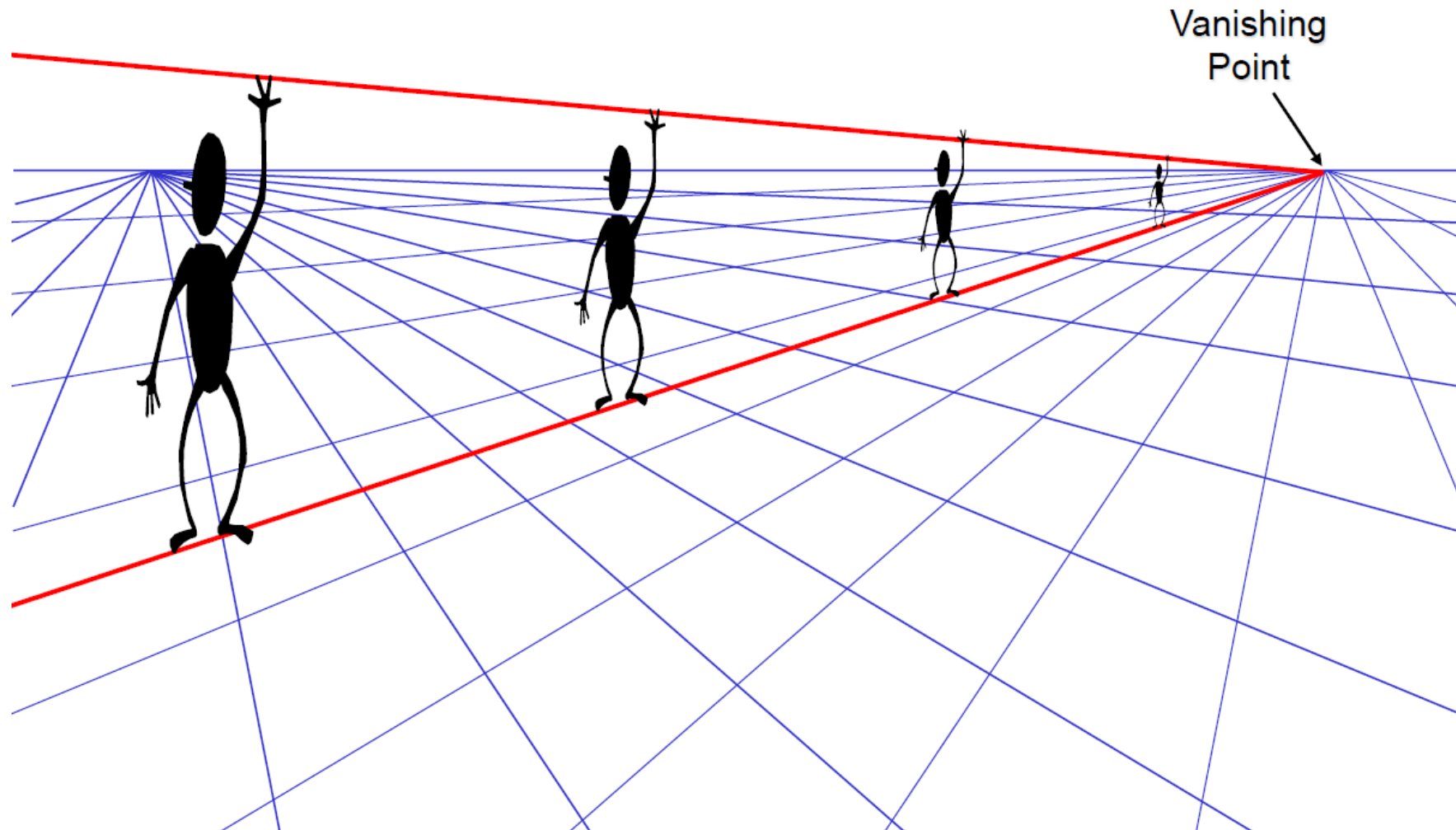


- The image lines of a set of parallel 3D lines meet at a “vanishing point”.
- Vanishing points of horizontal lines are located at the horizon

Depth Cues - Perspective

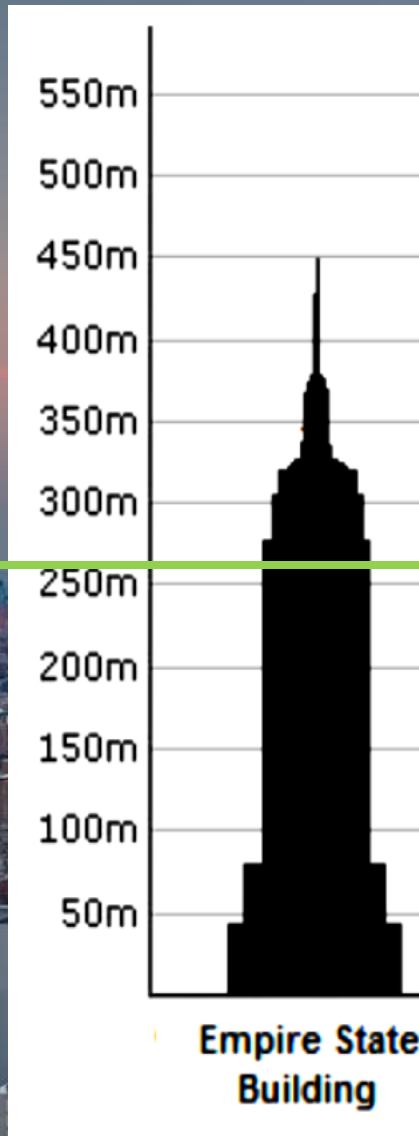


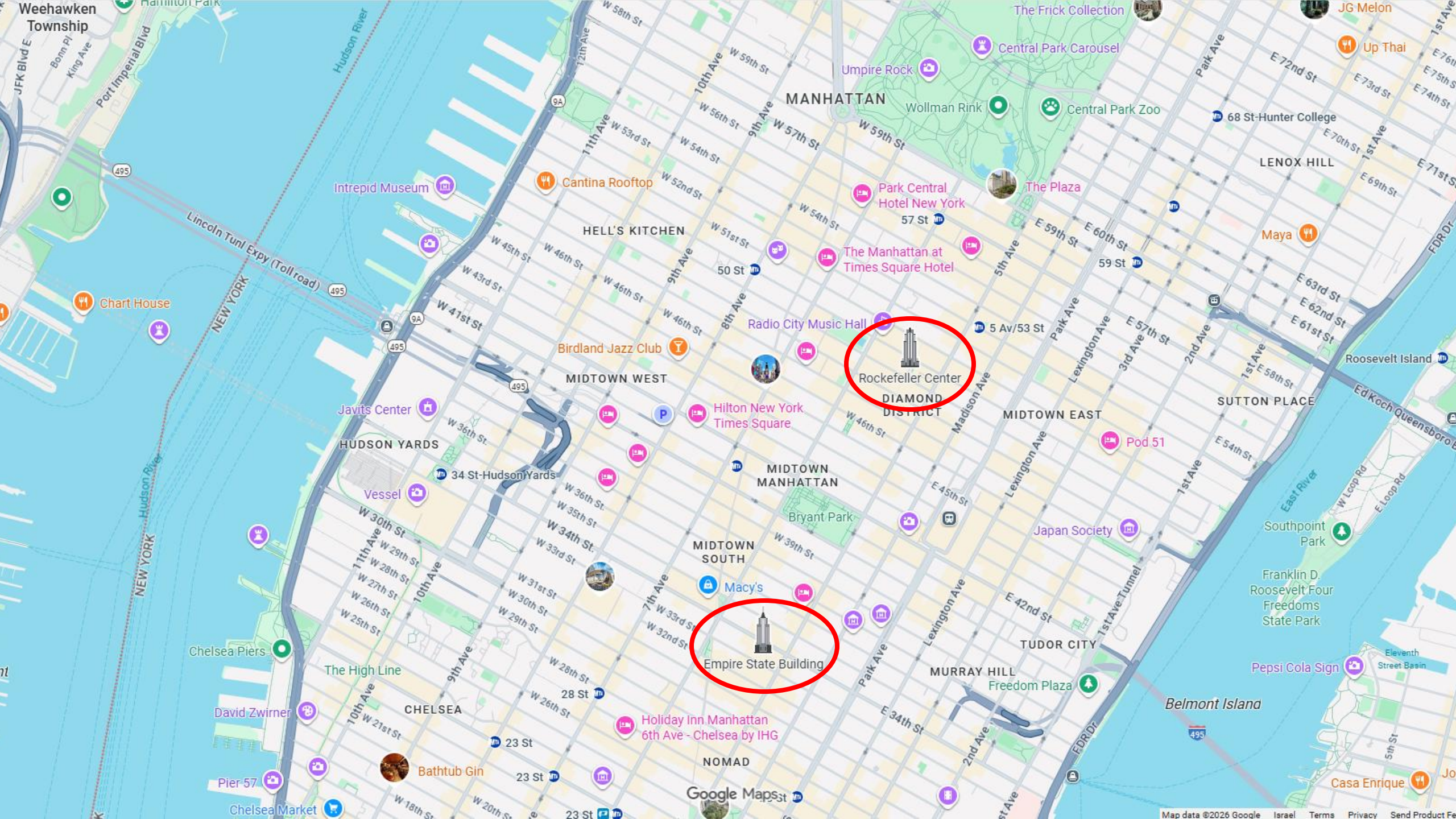
Vanishing Points



At what Elevation is the Camera?

Taken from Rockefeller Center Observation Deck, 260m





MANHATTAN

HELL'S KITCHEN

MIDTOWN WEST

MIDTOWN MANHATTAN

MIDTOWN SOUTH

CHELSEA

Rockefeller Center
DIAMOND DISTRICT

Empire State Building

Park Central
Hotel New York
57 St

The Manhattan at
Times Square Hotel

Radio City Music Hall

Hilton New York
Times Square

Macy's

Holiday Inn Manhattan
6th Ave - Chelsea by IHG

MIDTOWN EAST

Japan Society

TUDOR CITY

MURRAY HILL

Intrepid Museum

Cantina Rooftop

Birdland Jazz Club

Javits Center

HUDSON YARDS

34 St-Hudson Yards

Vessel

The High Line

David Zwirner

Pier 57

Chelsea Market

Bathtub Gin

Google Maps

Pod 51

5 Av/53 St

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

Lexington Ave

LENOX HILL

Maya

SUTTON PLACE

Southpoint Park

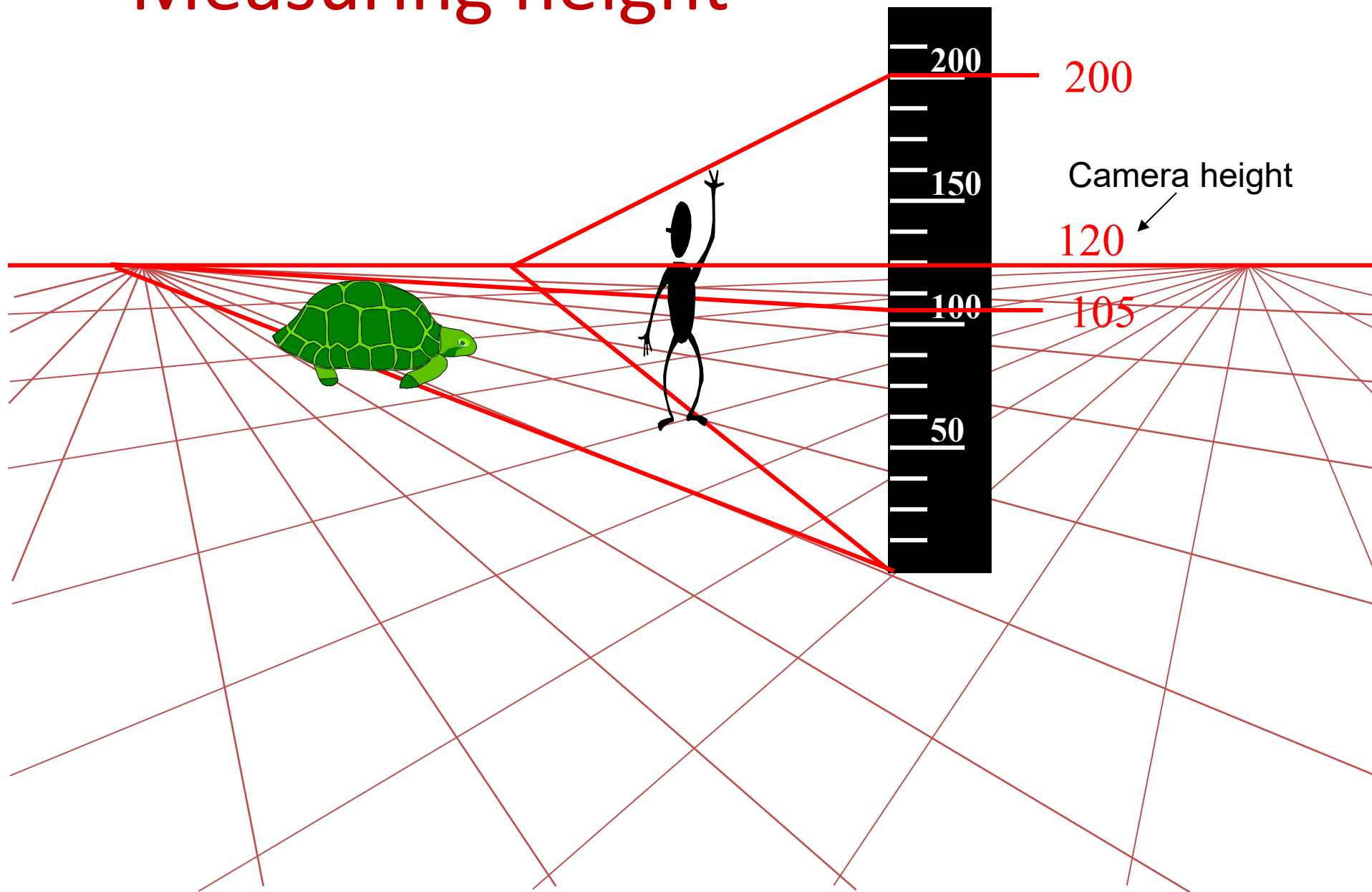
Franklin D. Roosevelt Four
Freedoms State Park

Pepsi Cola Sign

Belmont Island

Casa Enrique

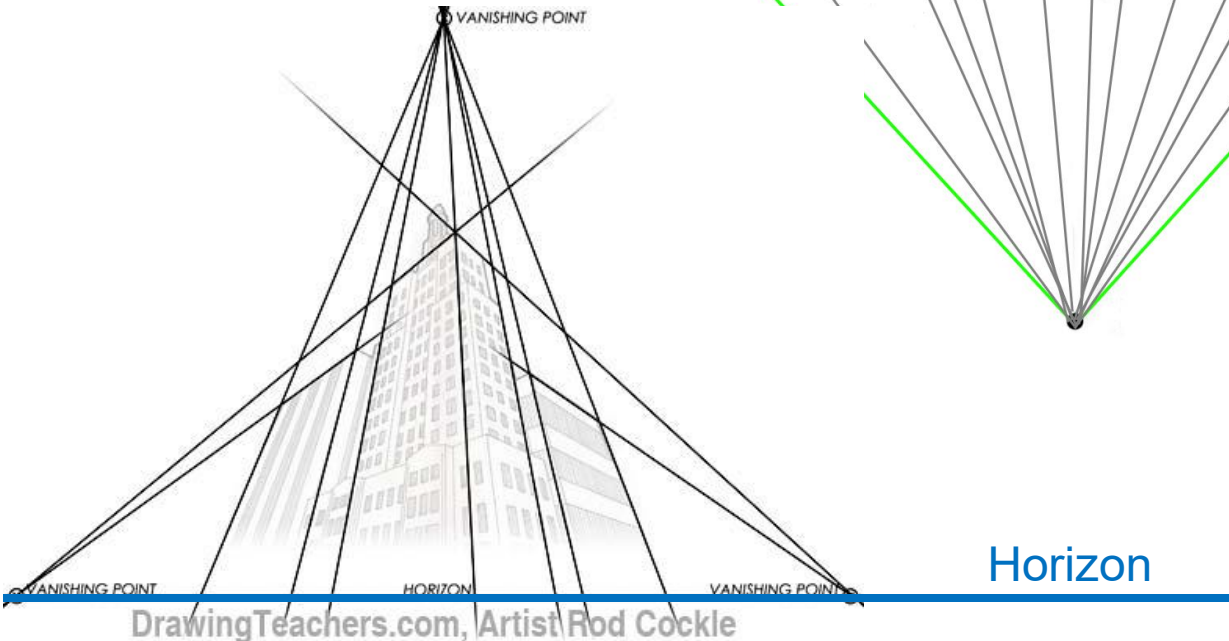
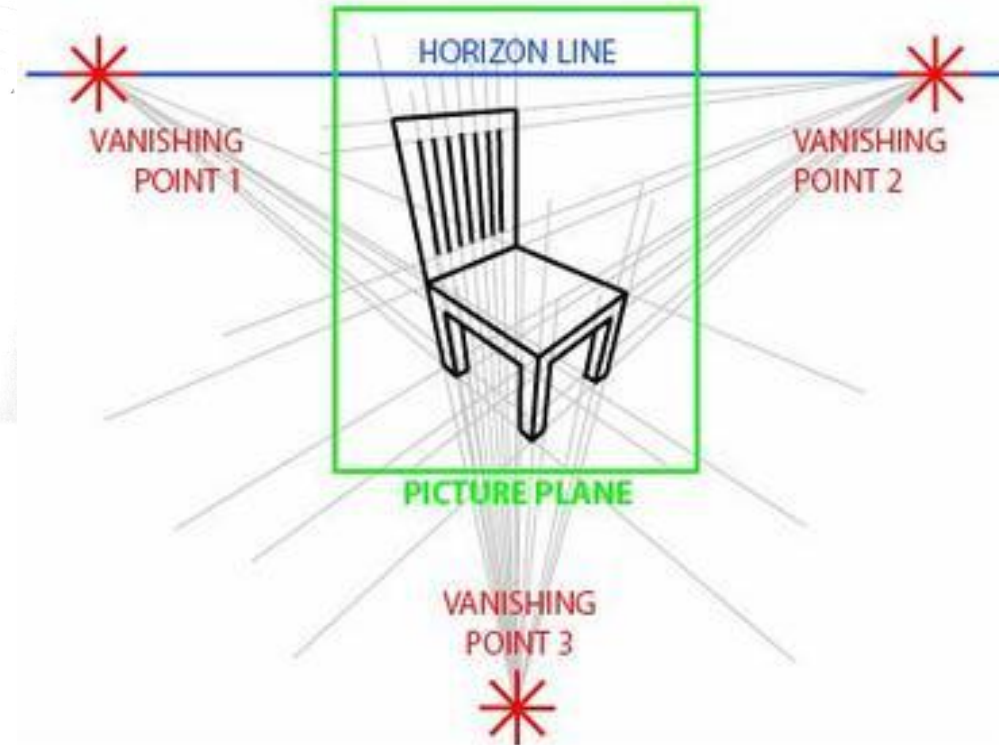
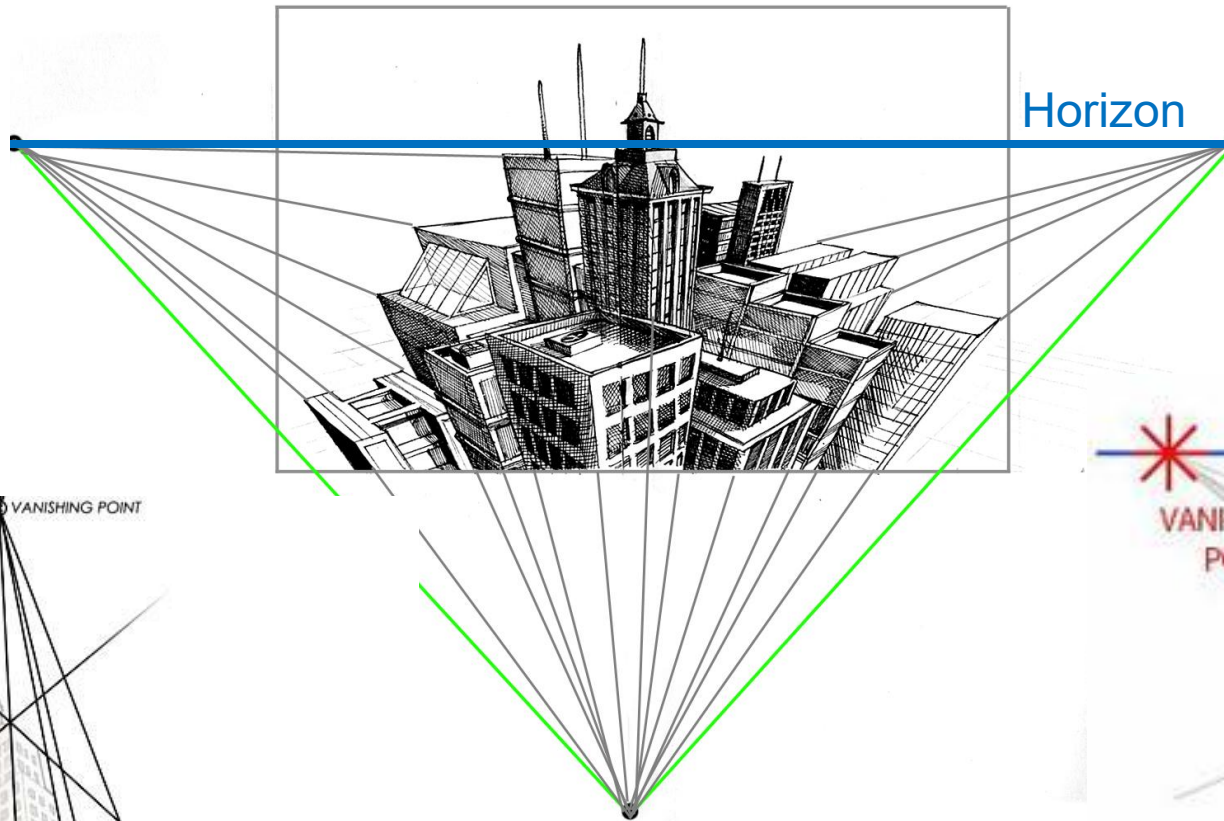
Measuring height



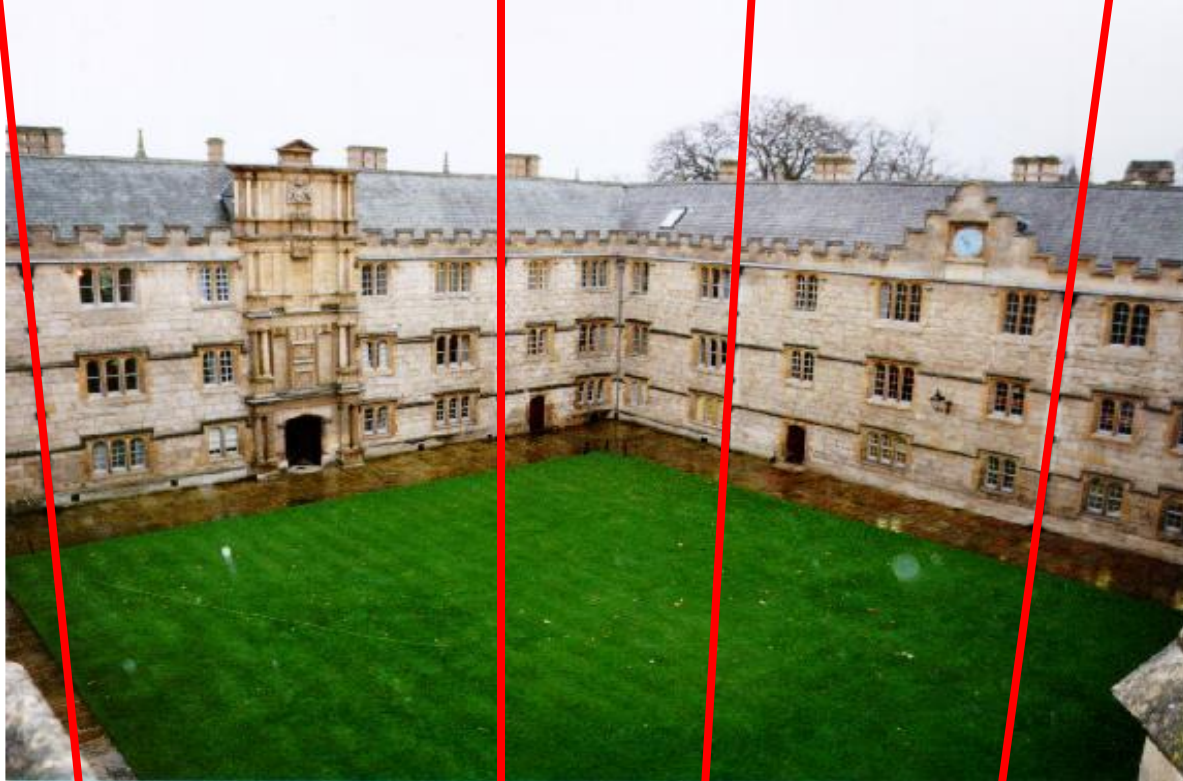
Is parachuter higher or lower than photographer



Three Point Perspective: Looking Up or Down

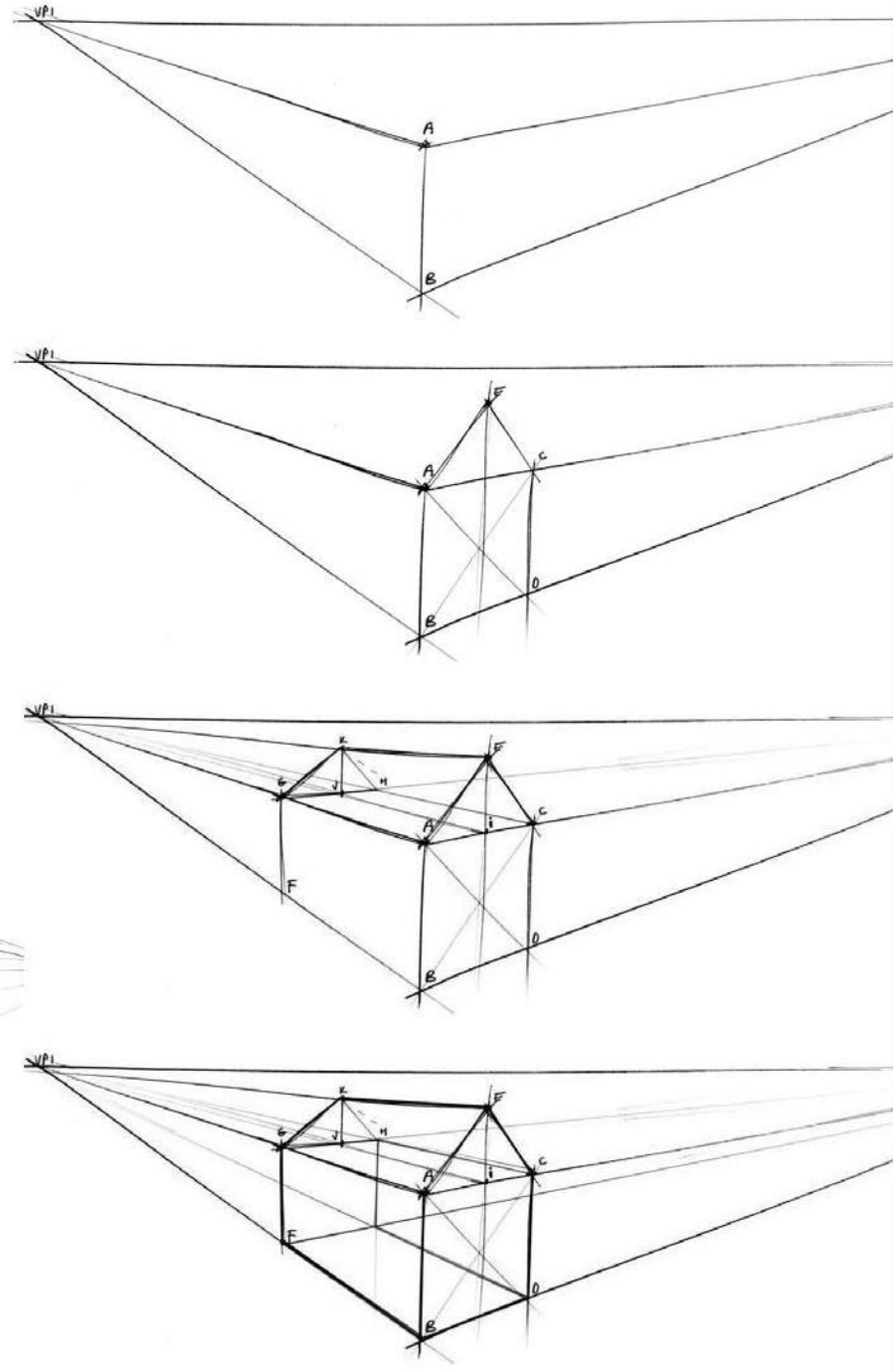


Vertical Vanishing Point



- All image lines corresponding to vertical lines meet at the Vertical Vanishing Point
- How can we identify vertical lines: Building lines, People, etc.

Drawing with Vanishing Points



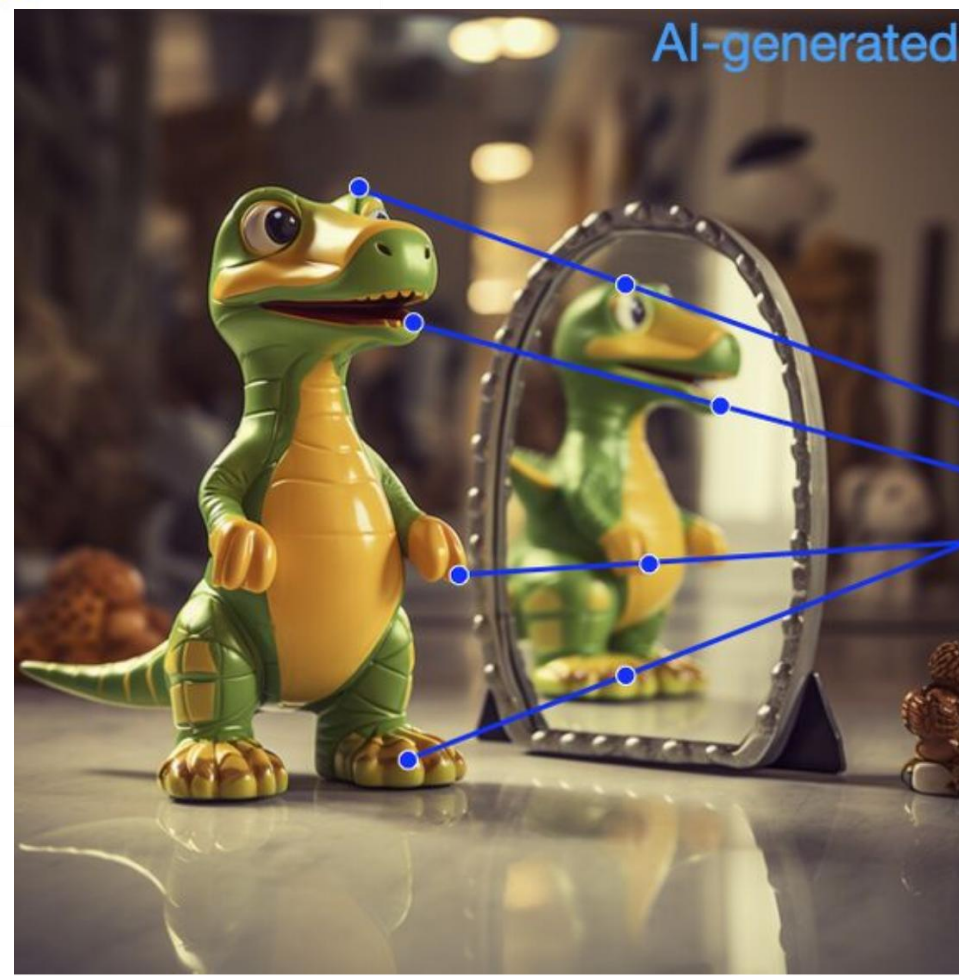
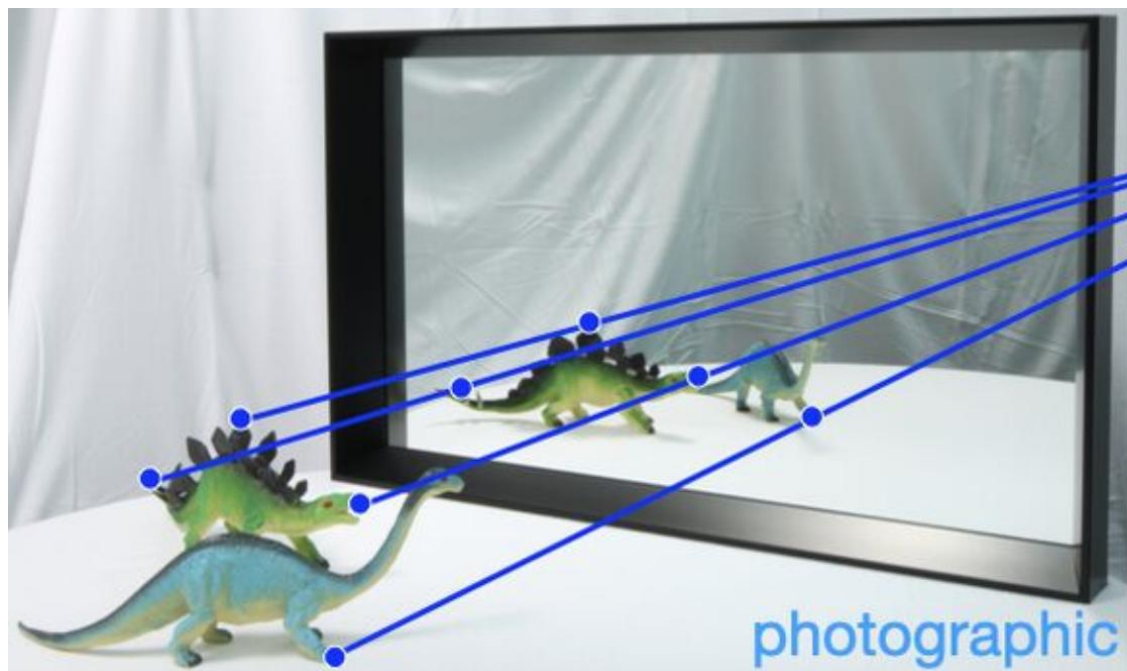
Art Before Perspective



Art Before Perspective



Vanishing Point Errors in old Generative AI



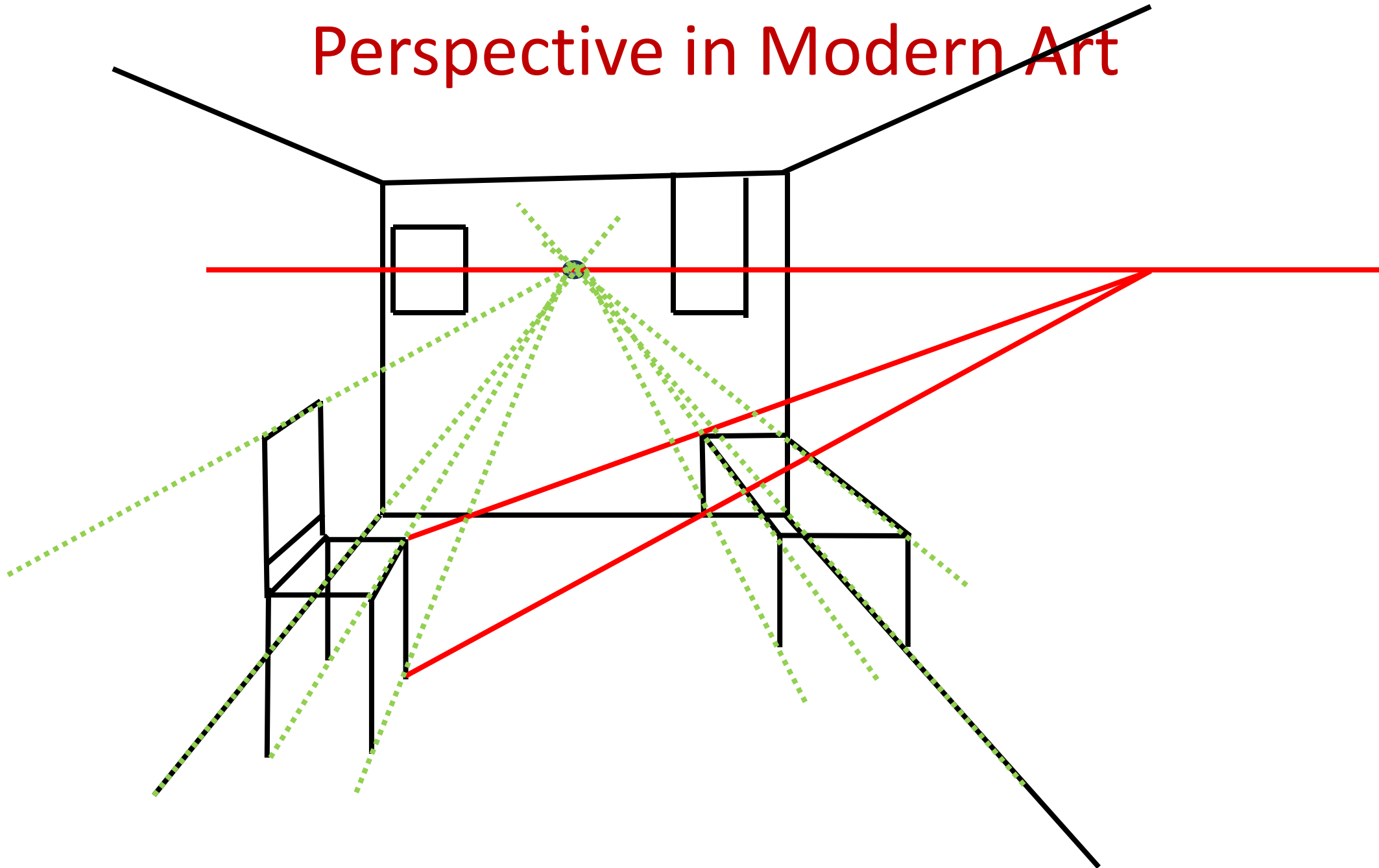
Straight Line Errors in old Generative AI



Perspective in Modern Art



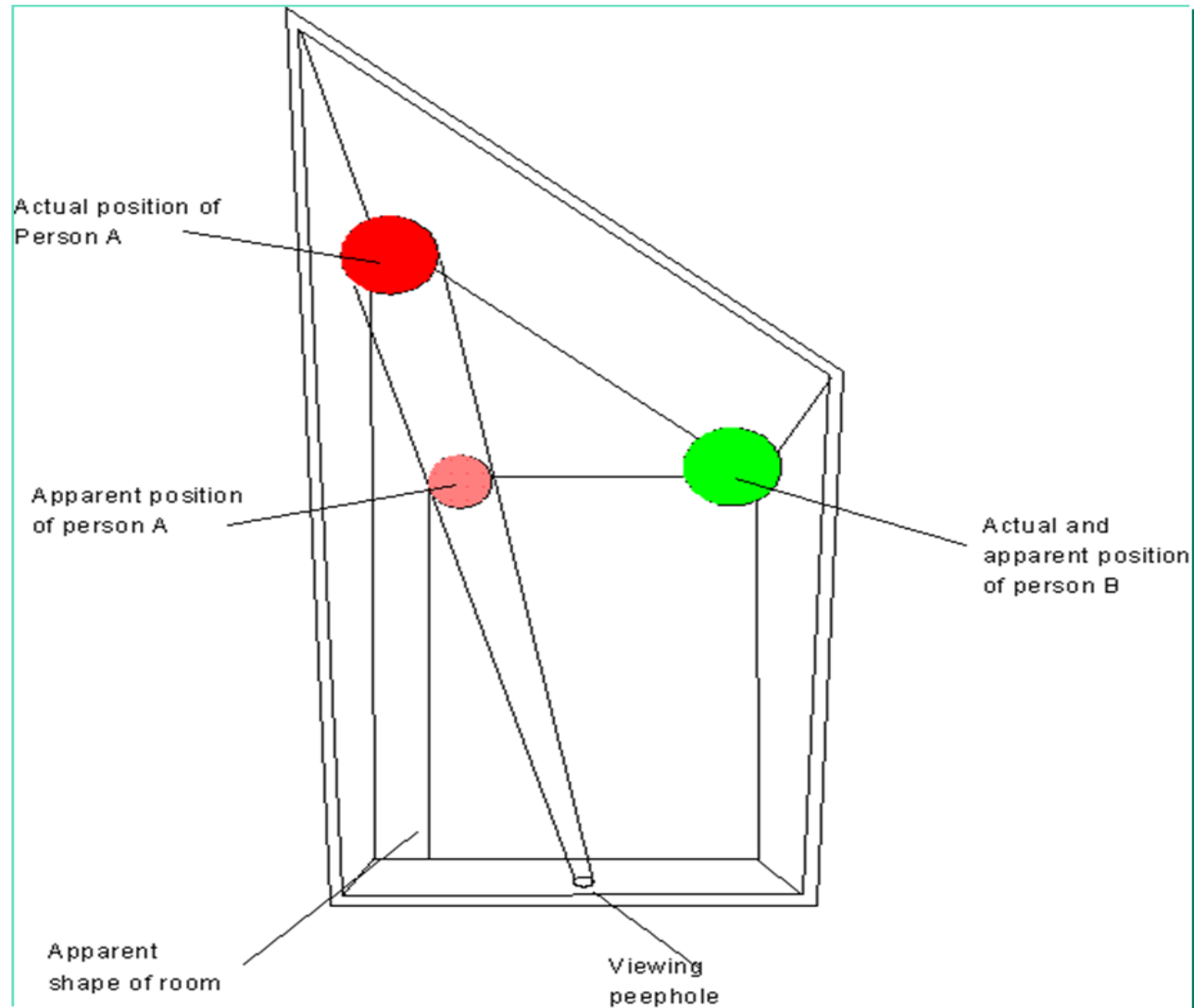
Perspective in Modern Art



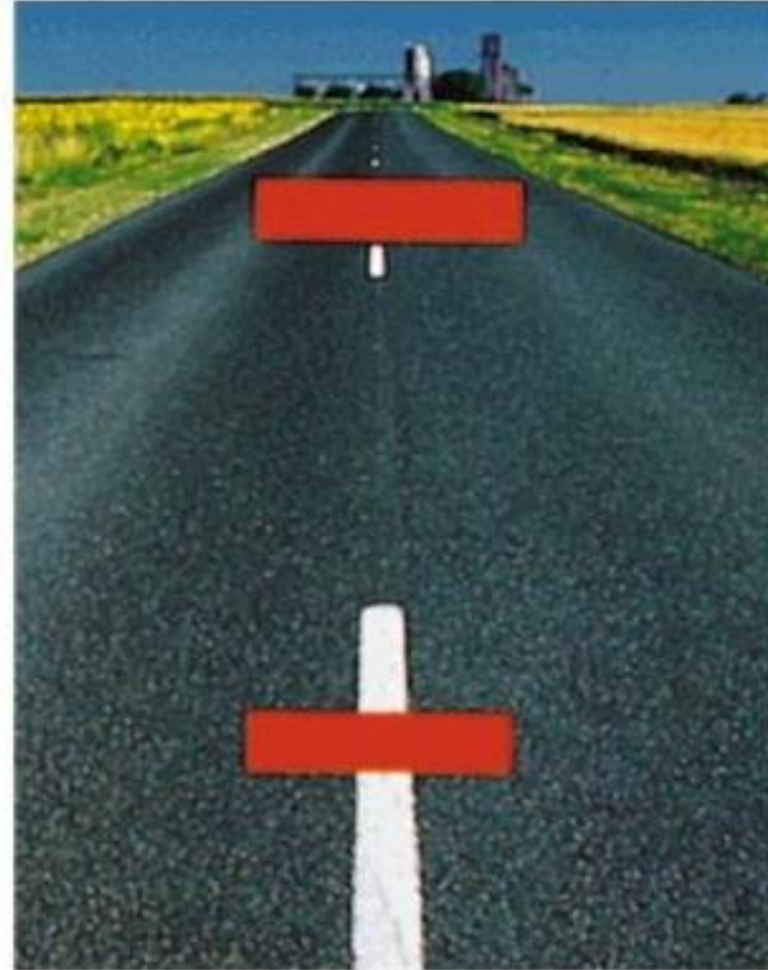
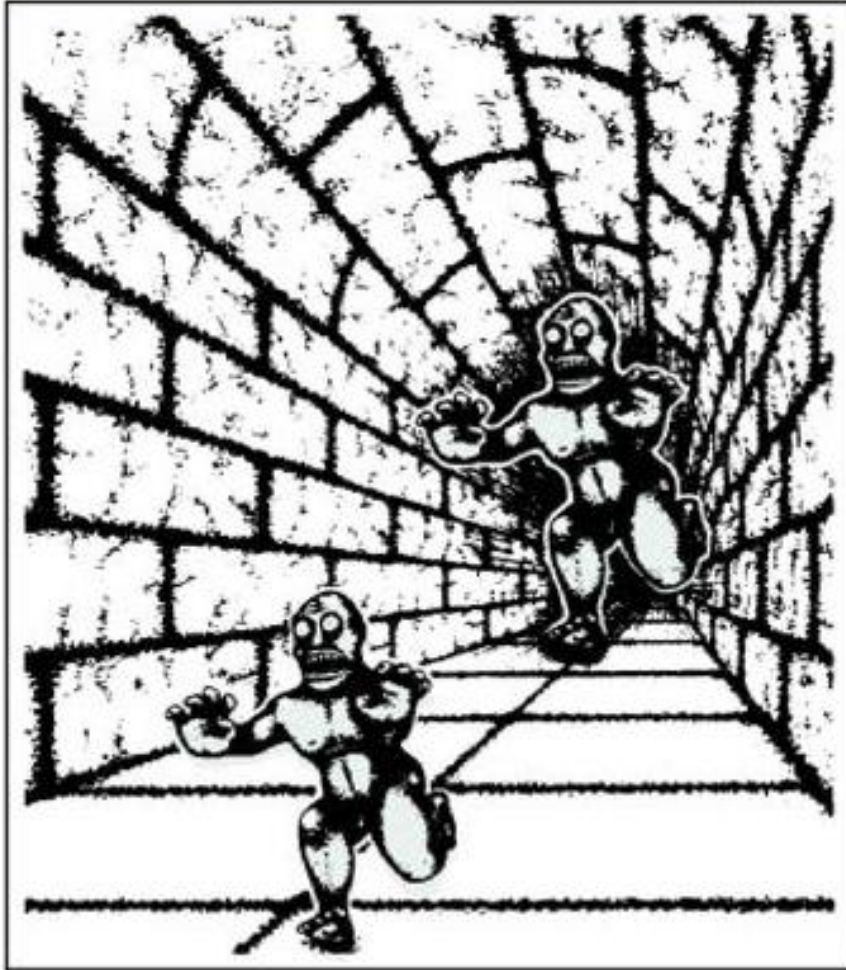
Visual Illusions



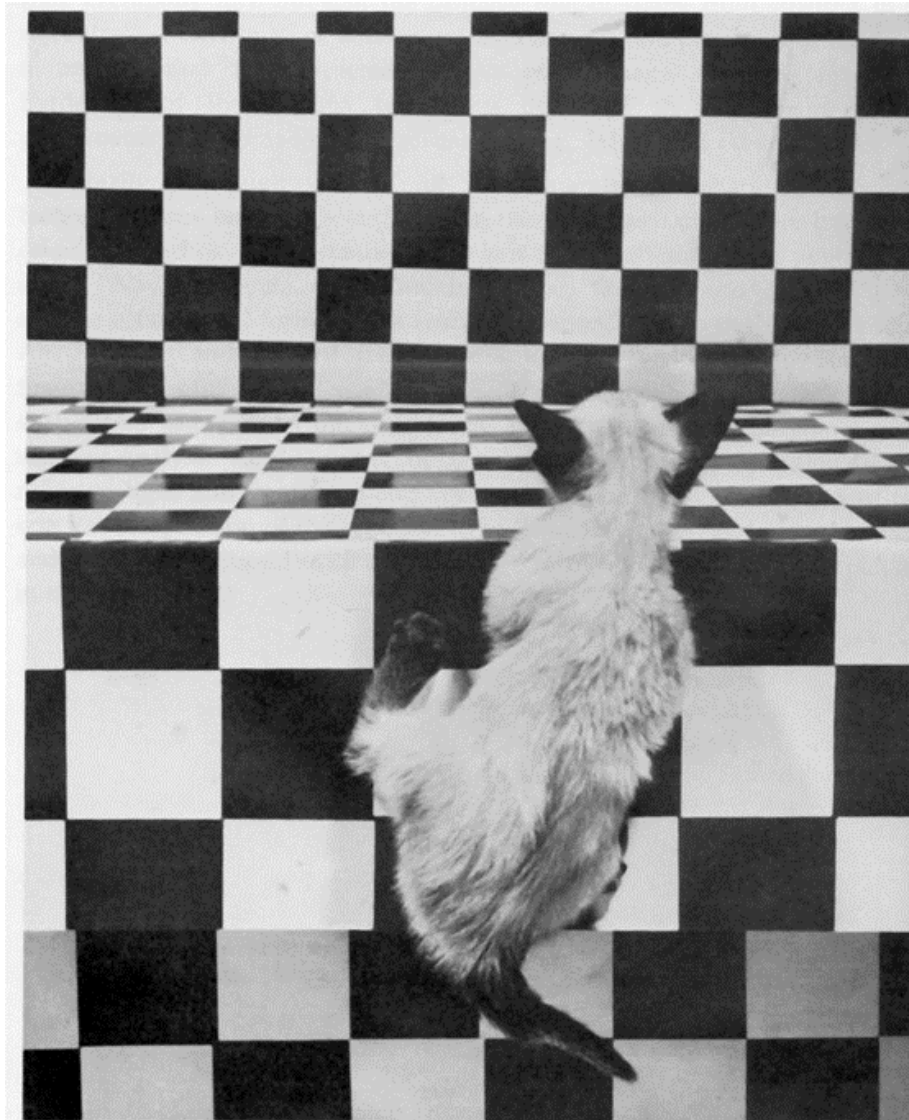
Ames Room



Visual Illusions



Visual Illusions

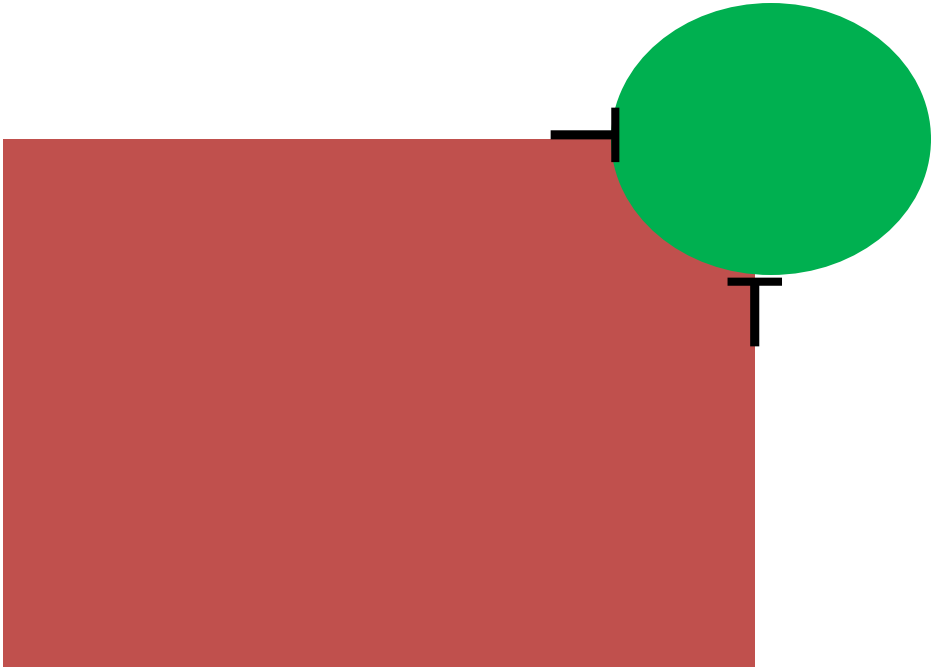


Visual Cues: Shape from X

- $X = :$
 - ☐ Occlusion
 - ☐ Shading
 - ☐ Texture
 - ☐ Focus
 - ☐ Motion
 - ☐ Haze
 - ☐ ...

- $X = :$
 - ☐ Highlights
 - ☐ Shadows
 - ☐ Silhouettes
 - ☐ Inter-reflections
 - ☐ Symmetry
 - ☐ Light Polarization
 - ☐ ...

Depth Cues - Occlusion



- Occluding object is closer than the occluded object
- Occlusion cue: T junctions
- Other cues: image size, location of bottom

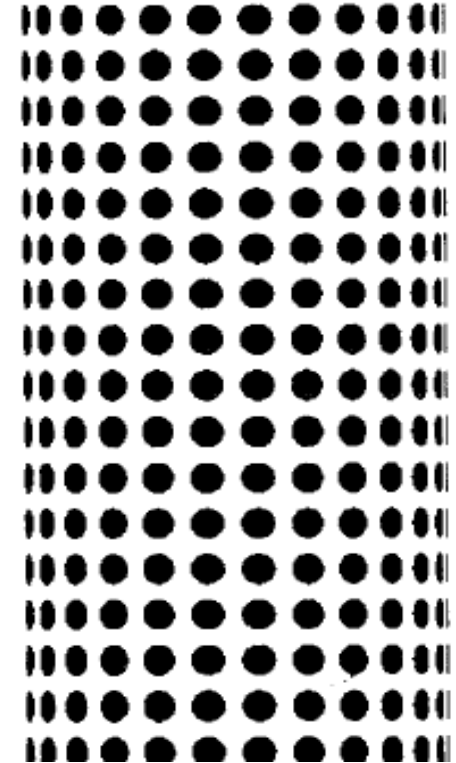
Depth Cues - Haze



Shadows



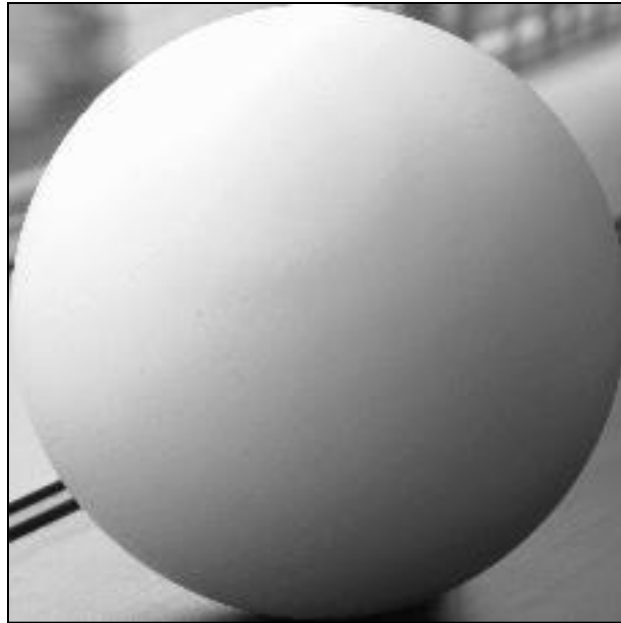
Texture Gradients



- Size and shape of texture elements – when identical (e.g. Kusama)
- Statistics of size and shape for randomly shaped elements

Photometric Stereo

Image Intensity and Geometry



Shading as a cue for shape reconstruction

What is the relation of intensity and shape?

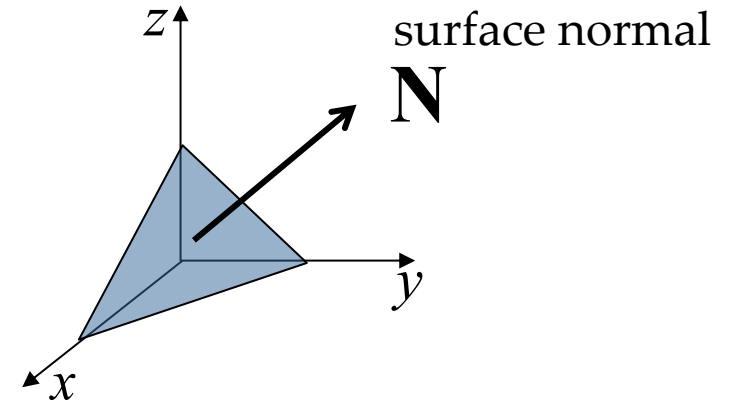
- Reflectance Map

Surface Normal

Equation of plane $Ax + By + Cz + D = 0$

or
$$z = -\left[\frac{A}{C}x + \frac{B}{C}y + \frac{D}{C}\right]$$

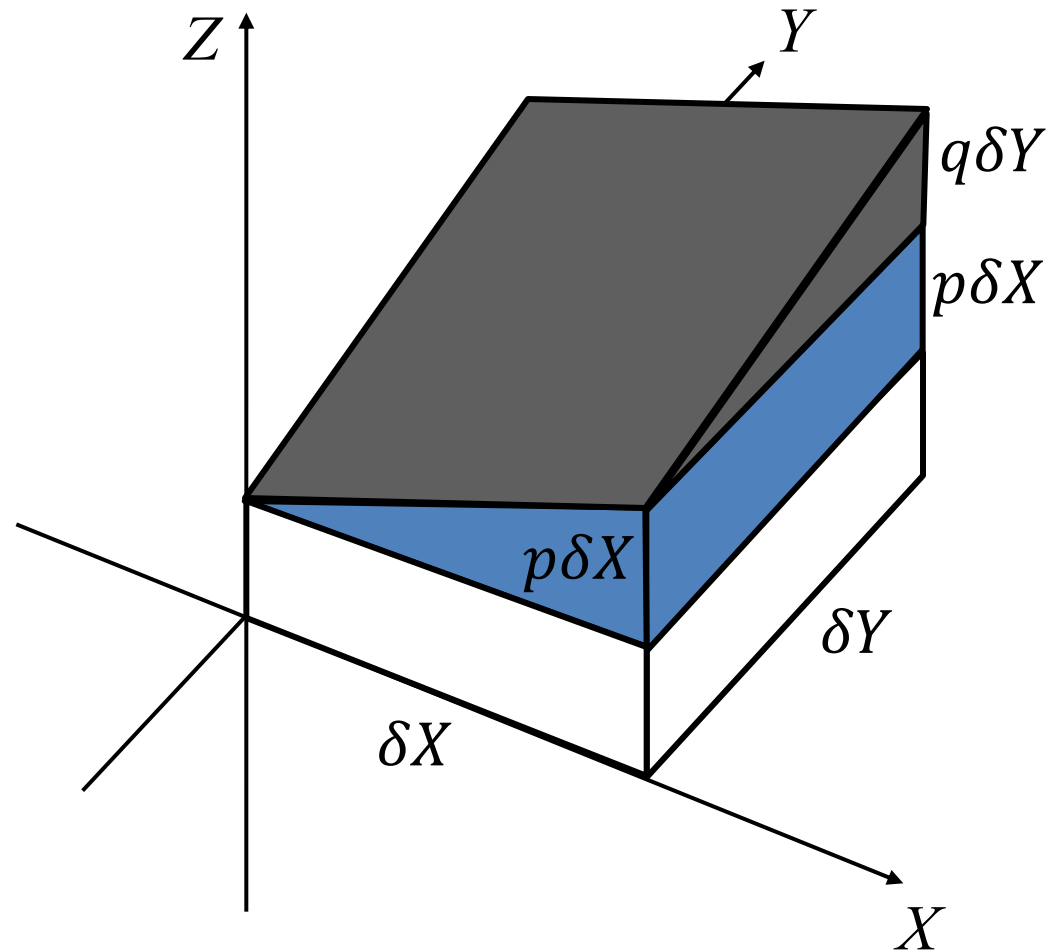
Let
$$-\frac{\partial z}{\partial x} = \frac{A}{C} = p \quad -\frac{\partial z}{\partial y} = \frac{B}{C} = q$$



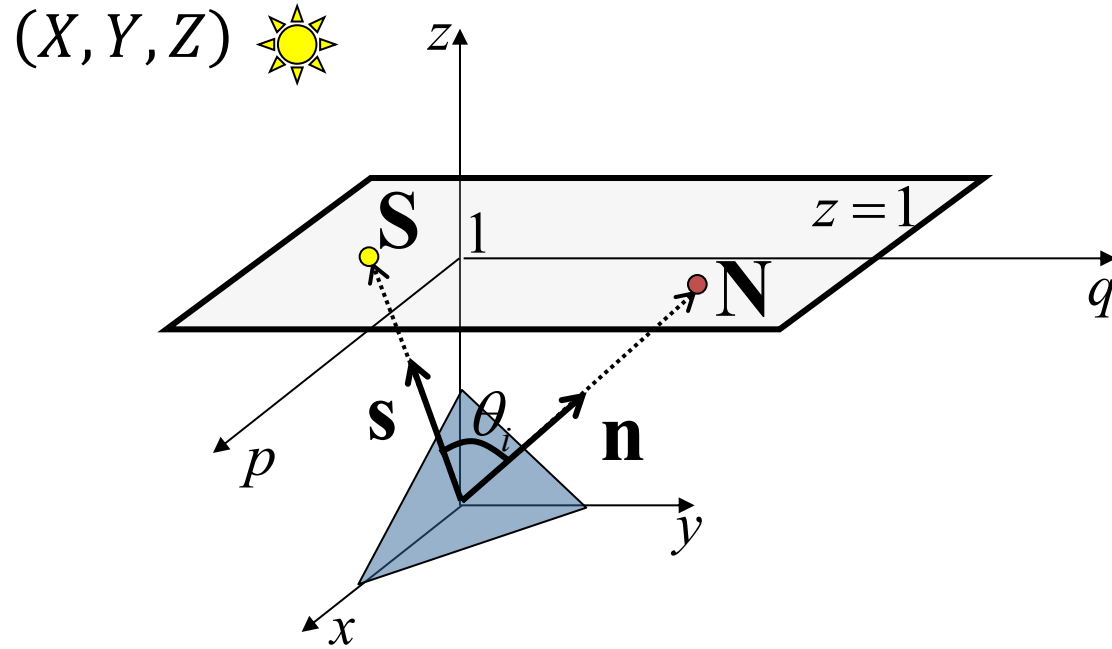
Surface normal

$$\mathbf{N} = \left(\frac{A}{C}, \frac{B}{C}, 1\right) = (p, q, 1)$$

Surface Normal



Gradient Space



Normal vector

$$\mathbf{n} = \frac{\mathbf{N}}{|\mathbf{N}|} = \frac{(p, q, 1)}{\sqrt{p^2 + q^2 + 1}}$$

Light Source vector

$$\mathbf{s} = \frac{\mathbf{S}}{|\mathbf{S}|} = \frac{(p_s, q_s, 1)}{\sqrt{p_s^2 + q_s^2 + 1}} \quad \mathbf{S} = \left(\frac{X}{Z}, \frac{Y}{Z}, 1 \right)$$

NN: Cosine Similarity

$$\cos \theta_i = \mathbf{n} \cdot \mathbf{s} = \frac{(pp_s + qq_s + 1)}{\sqrt{p^2 + q^2 + 1} \sqrt{p_s^2 + q_s^2 + 1}}$$

$z = 1$ plane is called the Gradient Space (pq plane)

- Every point in it corresponds to a particular surface orientation

Reflectance Map

Relates image irradiance $I(x,y)$ to surface orientation (p,q) for given light source direction and surface reflectance

Lambertian case (matte, diffuse)

k : light source brightness

ρ : surface albedo (reflectance)

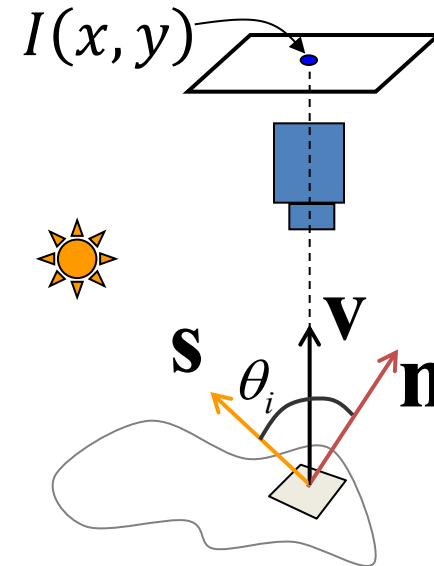
c : constant (optical system)

Image irradiance:

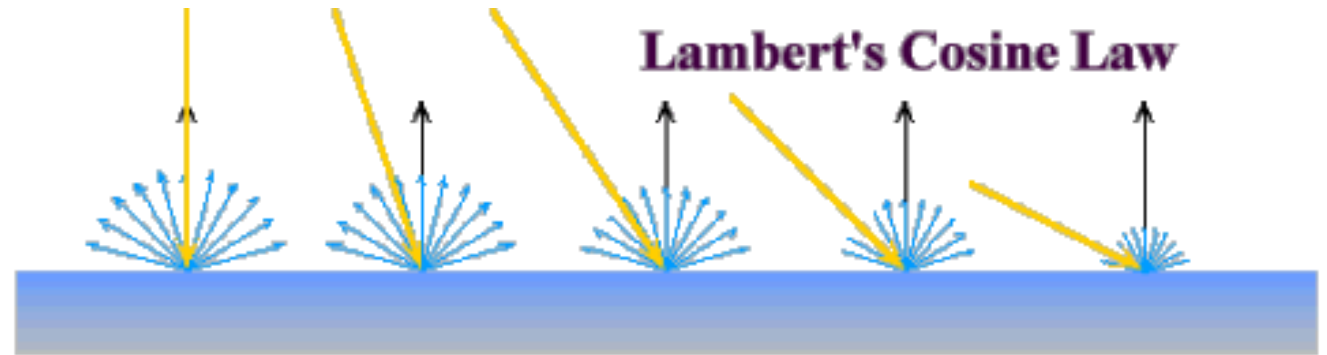
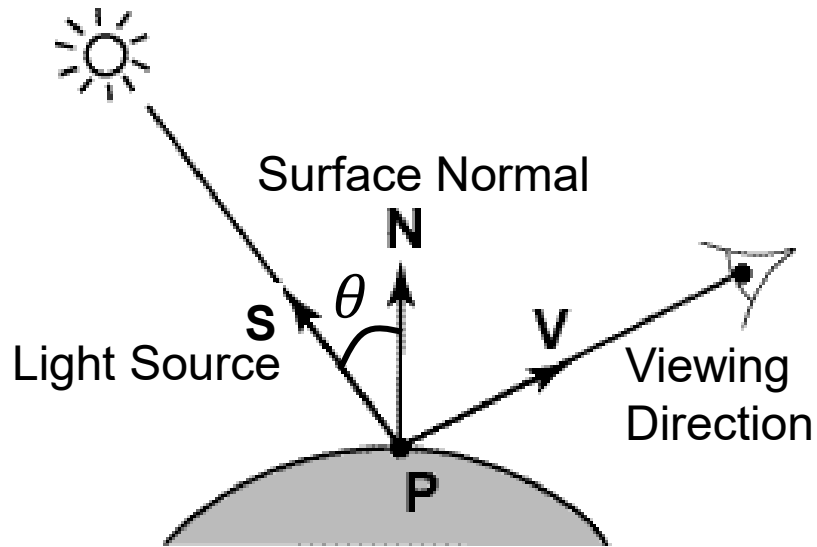
$$I = \rho k c \cos \theta_i = \rho k c \mathbf{n} \cdot \mathbf{s}$$

Let $\rho k c = 1$ then

$$I = \cos \theta_i = \mathbf{n} \cdot \mathbf{s}$$



Diffuse Reflection (Matte) “Lambertian”



- Visible Intensity: $I \approx \cos \theta$
- θ is angle between light source $\mathbf{S}$ and surface normal $\mathbf{N}$
- I does not depend on viewing direction $\mathbf{V}$
- In Shape from Shading, we compute surface normal map

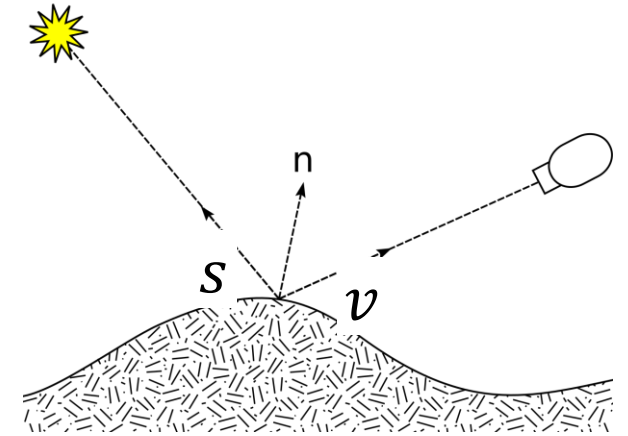
Reflectance Models

Lambertian Model: $I = k \cos \theta$

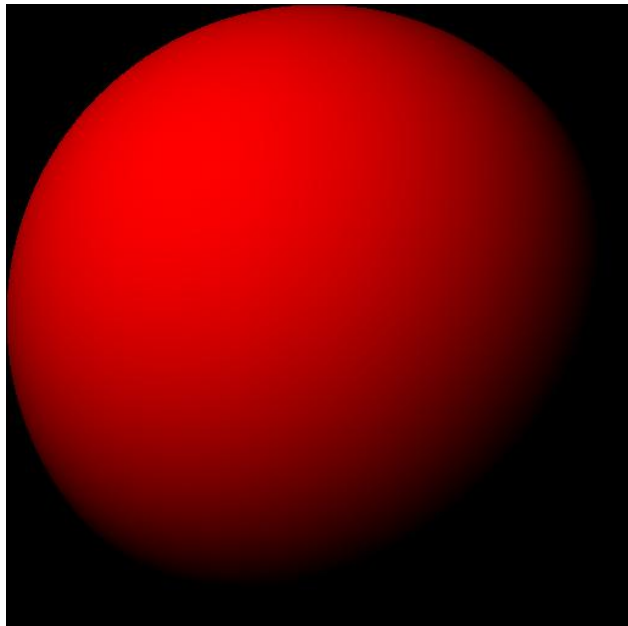
Most General: BRDF - Bidirectional reflectance distribution function

Let $\mathbf{n}=(0,0,1)$: BRDF is a function of 4 variables ($\mathbf{s}, \mathbf{v}$)

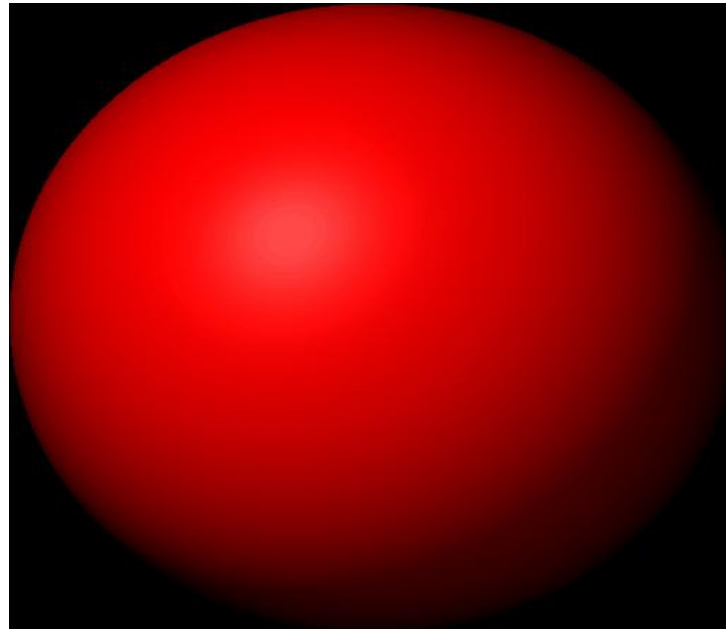
BRDF can be measured for every surface.



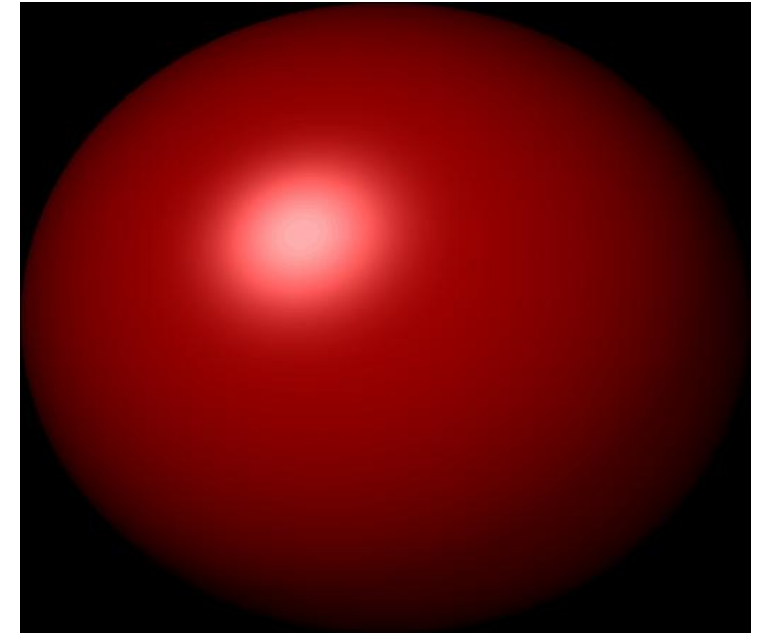
Lambertian albedo



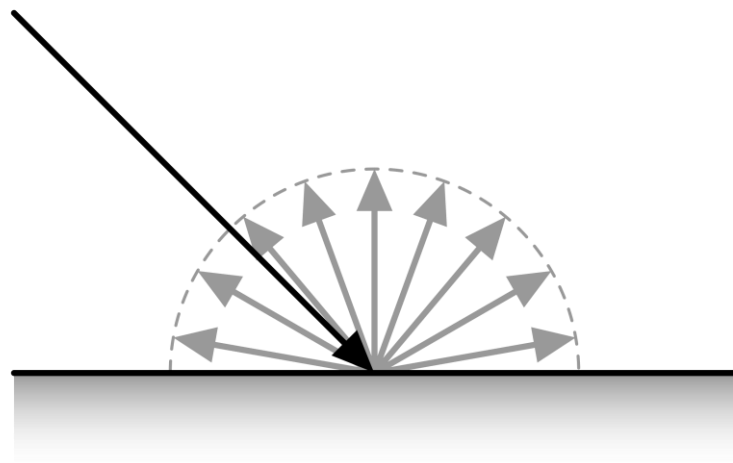
Little specularity



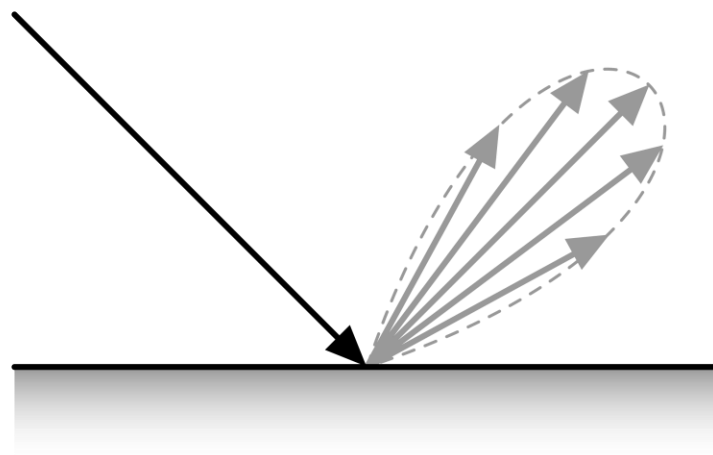
More specularity



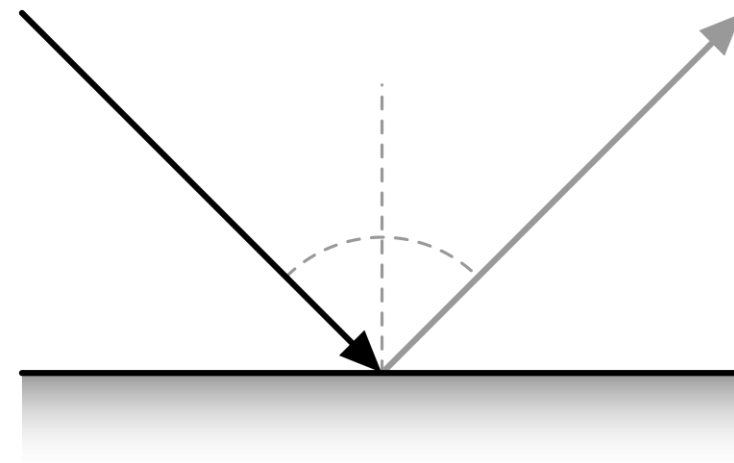
Simple Reflectance Models



Diffuse



Glossy



Mirror

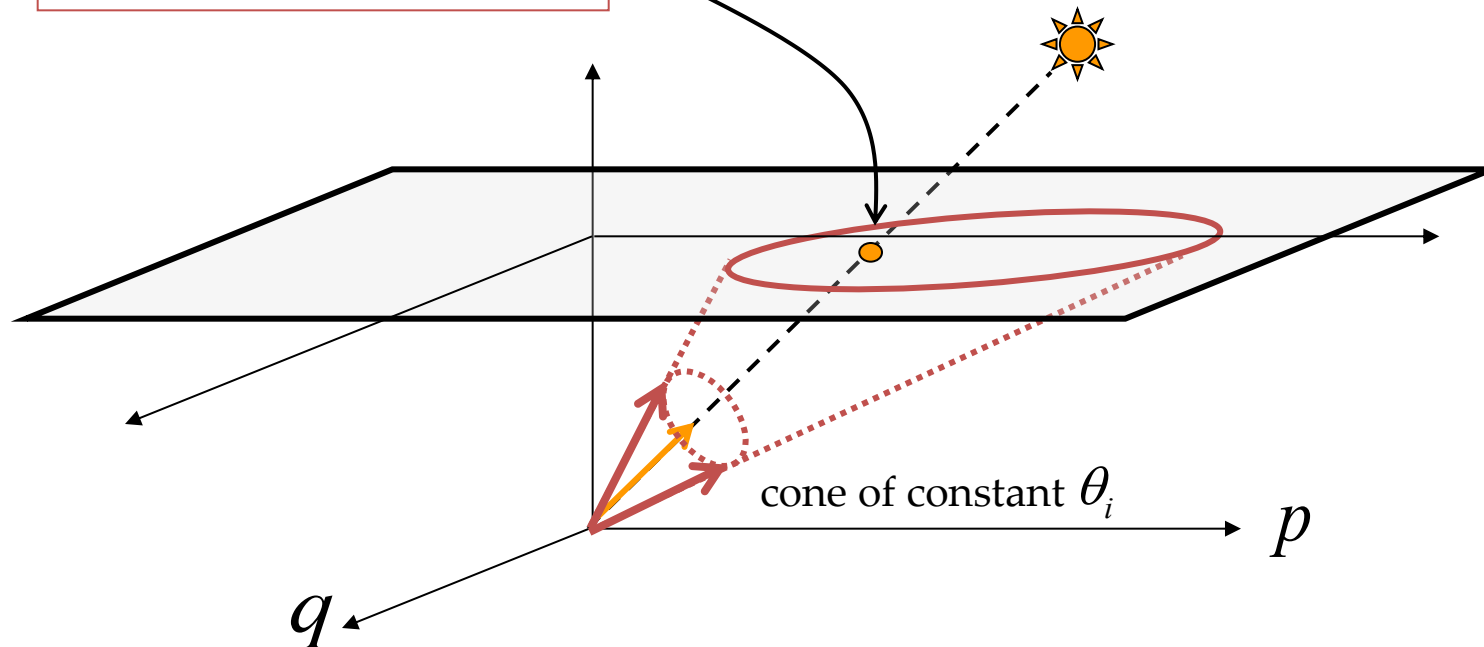
Lambertian Reflectance Map

Lambertian case

$$I = \cos \theta_i = \mathbf{n} \cdot \mathbf{s} = \frac{(pp_s + qq_s + 1)}{\sqrt{p^2 + q^2 + 1} \sqrt{p_s^2 + q_s^2 + 1}} = R(p, q)$$

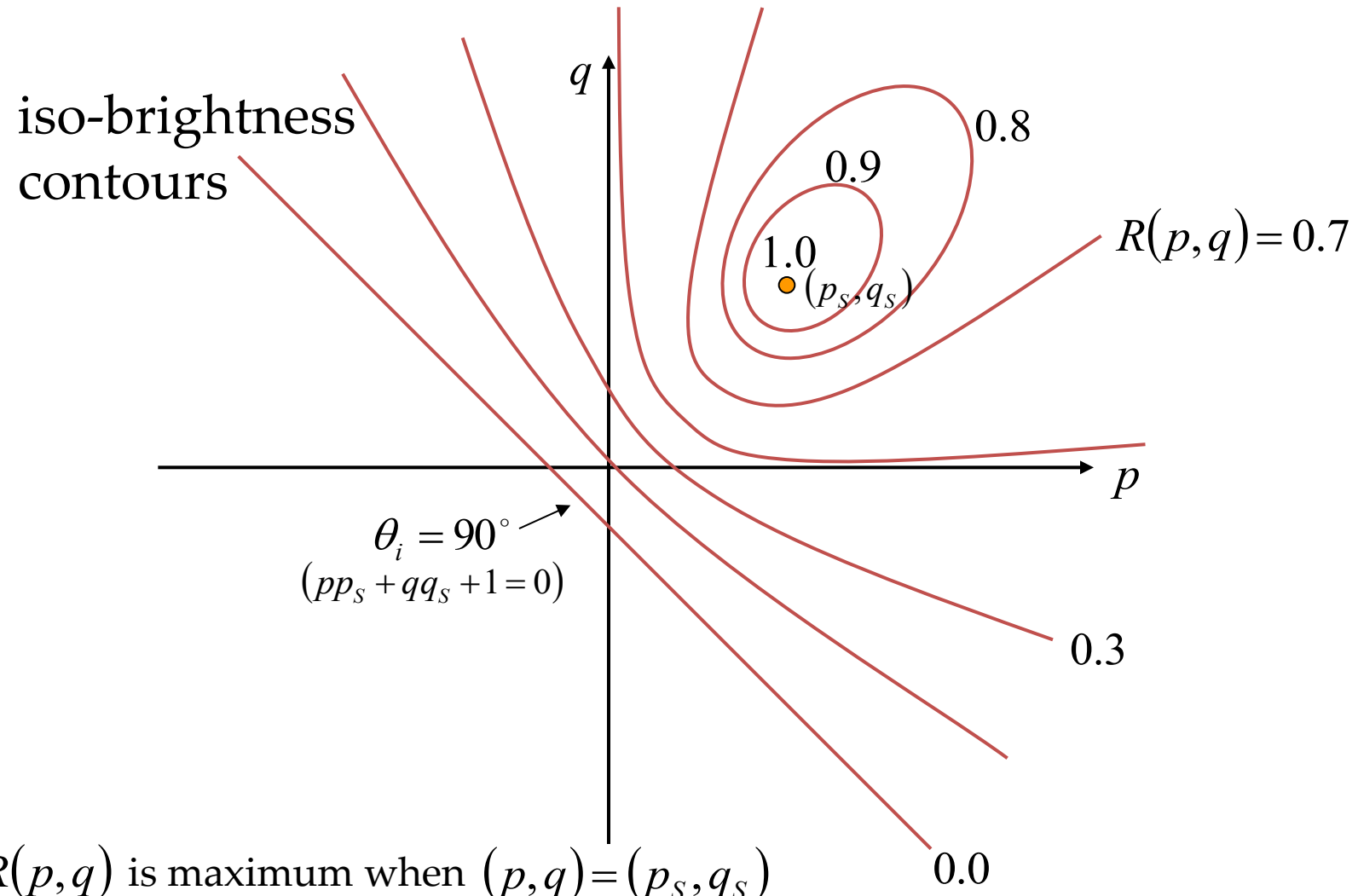
Reflectance Map (Lambertian)

Iso-brightness contour



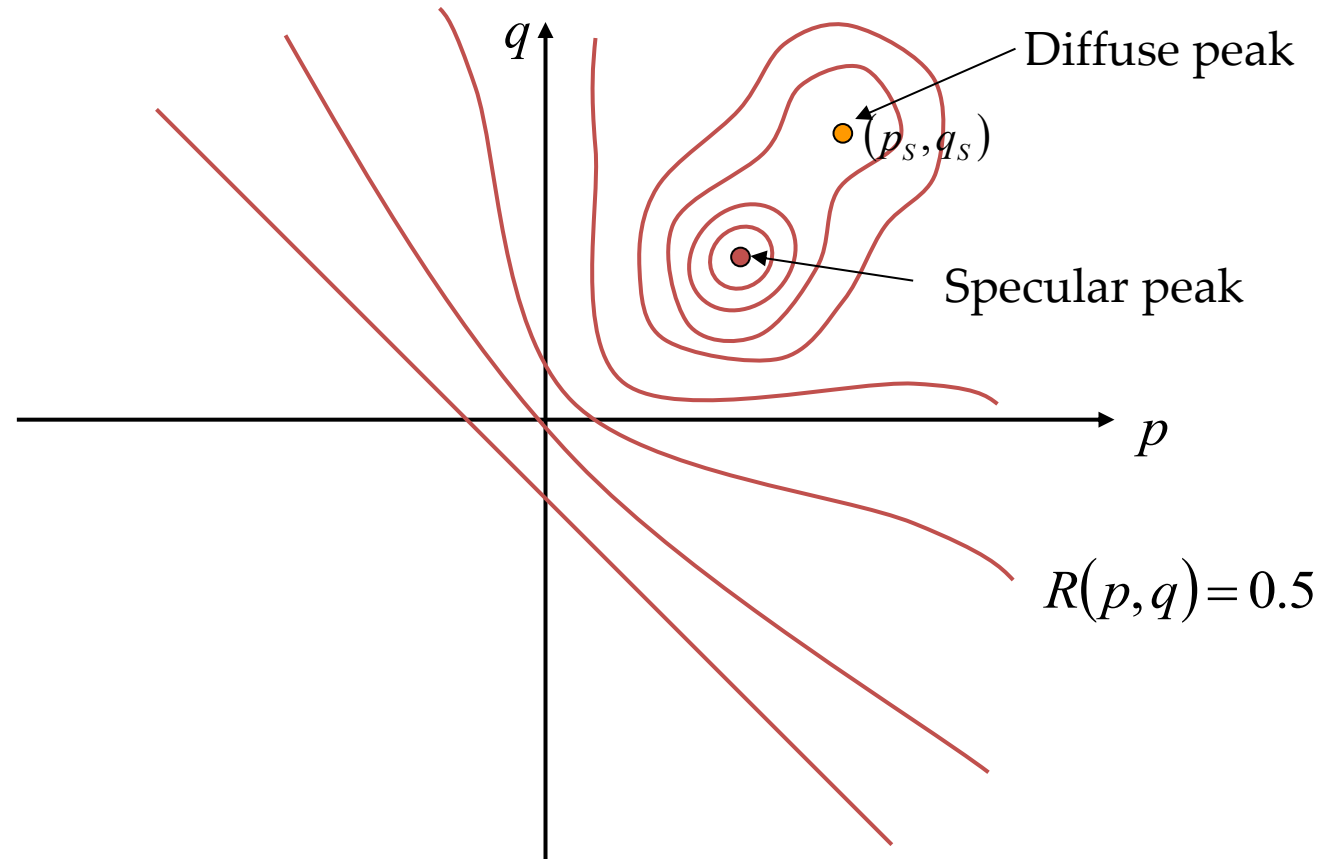
Lambertian Reflectance Map

Lambertian case



Reflectance Map

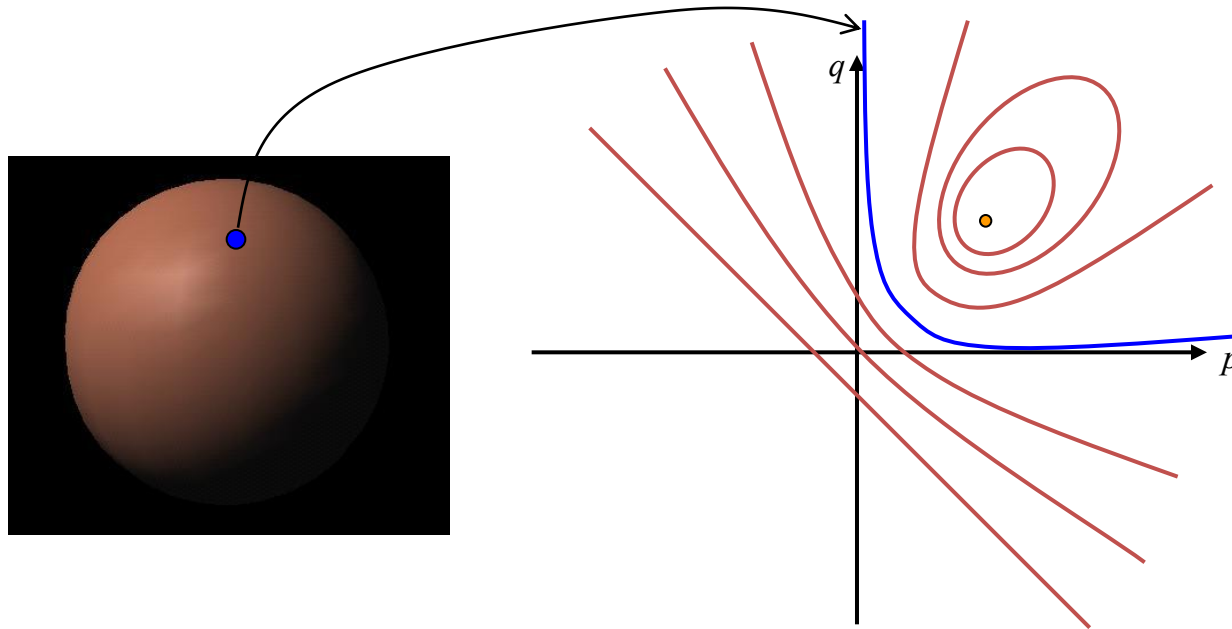
Glossy surfaces



Single-Image Shape Recovery?

Given a single image of an object with known surface reflectance taken under a known light source, can we recover the shape of the object?

Given $R(p, q)$ ((p_s, q_s) and surface reflectance) can we determine (p, q) uniquely for each image point?



NO

2 Approaches

Take more images with different light sources

- Photometric stereo

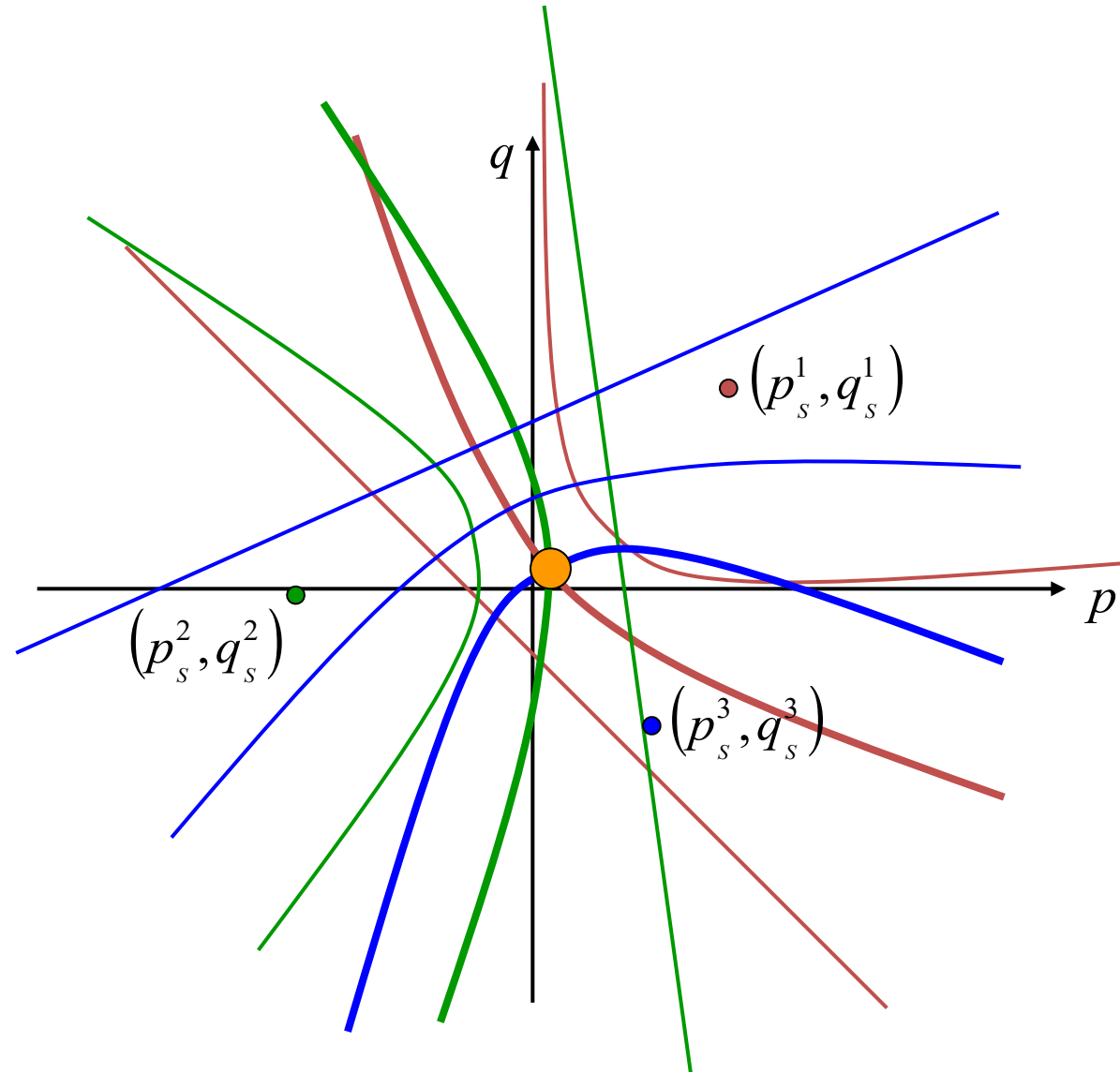
Add constraints

- Shape-from-shading

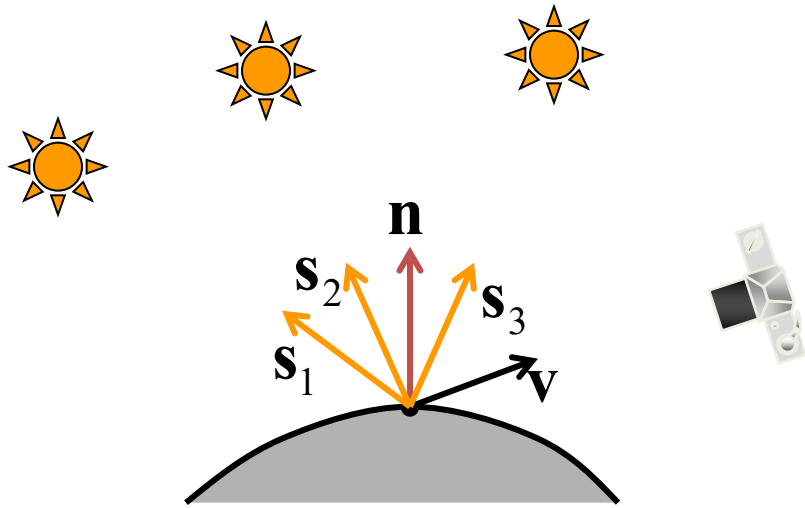
Photometric Stereo

Take multiple images, each from a different light source direction

Based on the intensity
at each image point as
captured in each image:



Photometric Stereo



Lambertian :

$$I = \rho kc \cos \theta_i = \rho \mathbf{n} \cdot \mathbf{s} \quad (kc = 1)$$

Image irradiance:

$$I_1 = \rho \mathbf{n} \cdot \mathbf{s}_1$$

$$I_2 = \rho \mathbf{n} \cdot \mathbf{s}_2$$

$$I_3 = \rho \mathbf{n} \cdot \mathbf{s}_3$$

- Can write this as a matrix equation:

$$\begin{bmatrix} I_1 \\ I_2 \\ I_2 \end{bmatrix} = \rho \begin{bmatrix} \mathbf{s}_1^T \\ \mathbf{s}_2^T \\ \mathbf{s}_3^T \end{bmatrix} \mathbf{n}$$

Solving the Equations

$$\underbrace{\begin{bmatrix} I_1 \\ I_2 \\ I_2 \end{bmatrix}}_{\substack{\mathbf{I} \\ 3 \times 1}} = \underbrace{\begin{bmatrix} \mathbf{s}_1^T \\ \mathbf{s}_2^T \\ \mathbf{s}_3^T \end{bmatrix}}_{\substack{\mathbf{S} \\ 3 \times 3}} \underbrace{\rho \mathbf{n}}_{\substack{\tilde{\mathbf{n}} \\ 3 \times 1}}$$

$$\tilde{\mathbf{n}} = \mathbf{S}^{-1} \mathbf{I}$$

← inverse

$$\rho = |\tilde{\mathbf{n}}|$$

$$\mathbf{n} = \frac{\tilde{\mathbf{n}}}{|\tilde{\mathbf{n}}|} = \frac{\tilde{\mathbf{n}}}{\rho}$$

More than Three Lights

Get better results by using more lights

$$\begin{bmatrix} I_1 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} \mathbf{s}_1^T \\ \vdots \\ \mathbf{s}_N^T \end{bmatrix} \rho \mathbf{n}$$

- Least squares solution:

$$\mathbf{I} = \mathbf{S} \tilde{\mathbf{n}} \quad \longleftarrow \quad N \times 1 = (\underline{N \times 3})(3 \times 1)$$

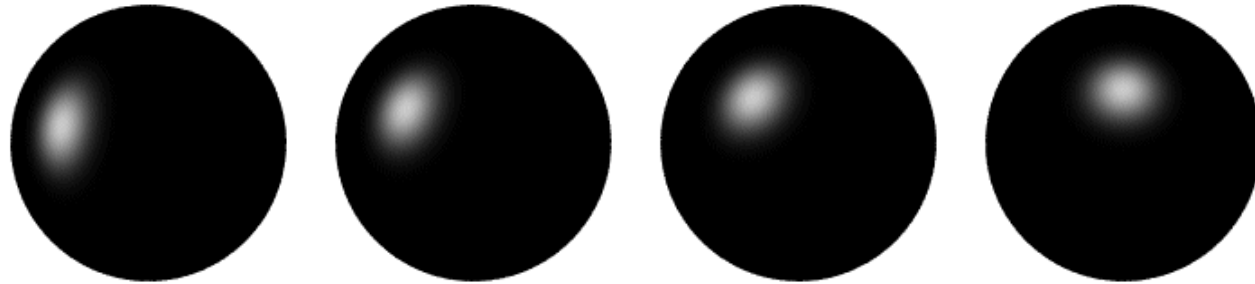
$$\mathbf{S}^T \mathbf{I} = \mathbf{S}^T \mathbf{S} \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{n}} = (\mathbf{S}^T \mathbf{S})^{-1} \mathbf{S}^T \mathbf{I}$$

- Solve for $\rho, \mathbf{n}$ as before Moore-Penrose pseudo inverse

Computing light source directions

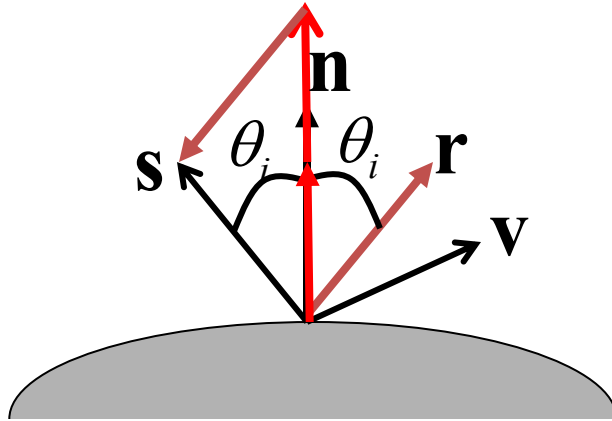
Trick: place a chrome sphere in the scene



– the location of the highlight tells you where the light source is

Recall the Rule for Specular Reflection

- For a perfect mirror, light is reflected about **N**

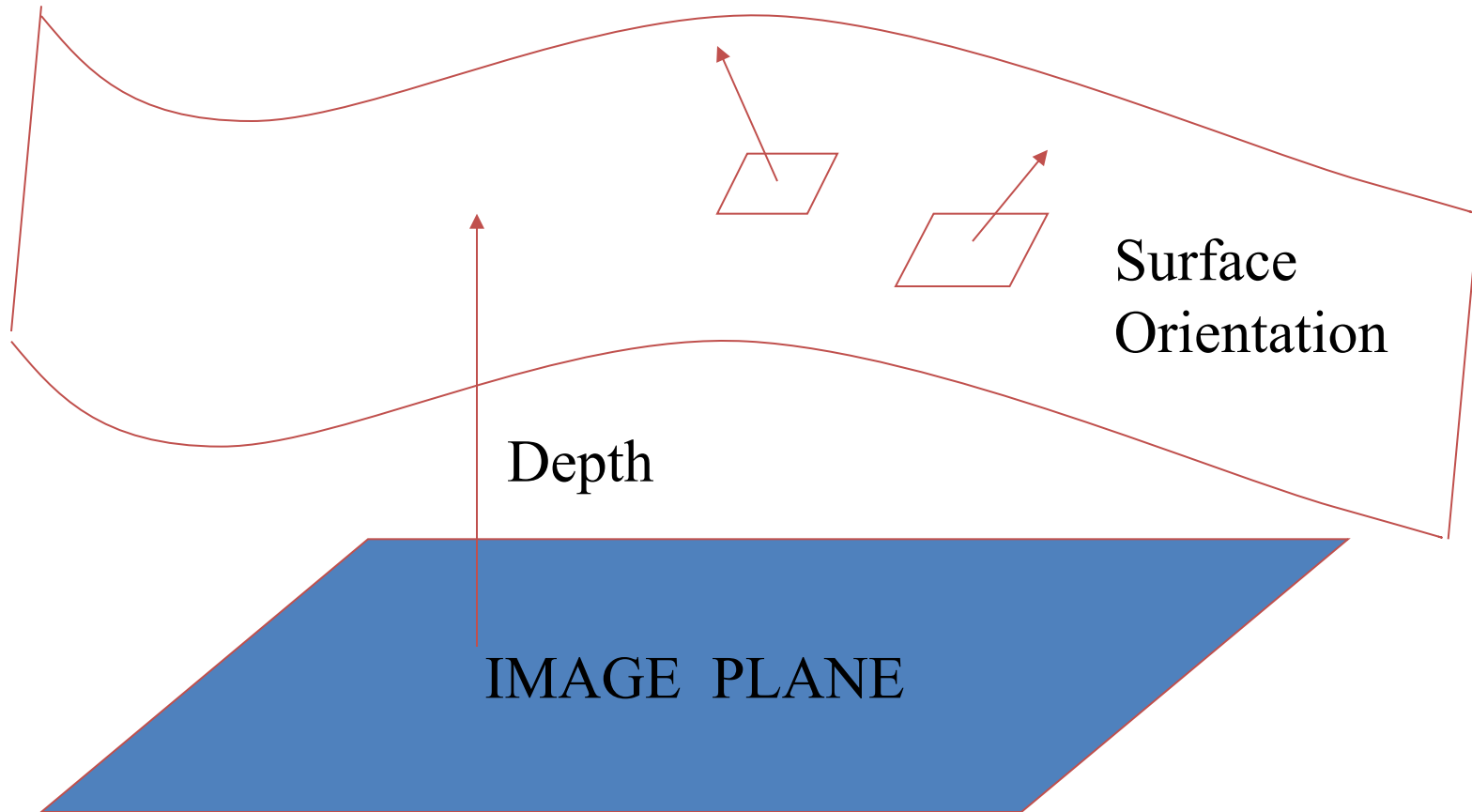


$$R_e = \begin{cases} R_i & \text{if } \mathbf{v} = \mathbf{r} \\ 0 & \text{otherwise} \end{cases}$$

- We see a highlight when $\mathbf{v} = \mathbf{r}$
- then $\mathbf{s}$ is given as follows:

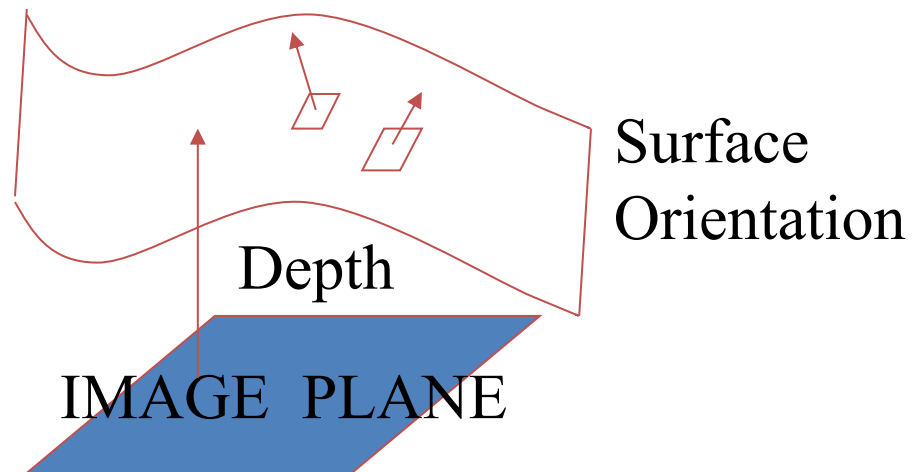
$$\mathbf{s} = 2(\mathbf{n} \cdot \mathbf{r})\mathbf{n} - \mathbf{r}$$

Normal Map vs. Depth Map



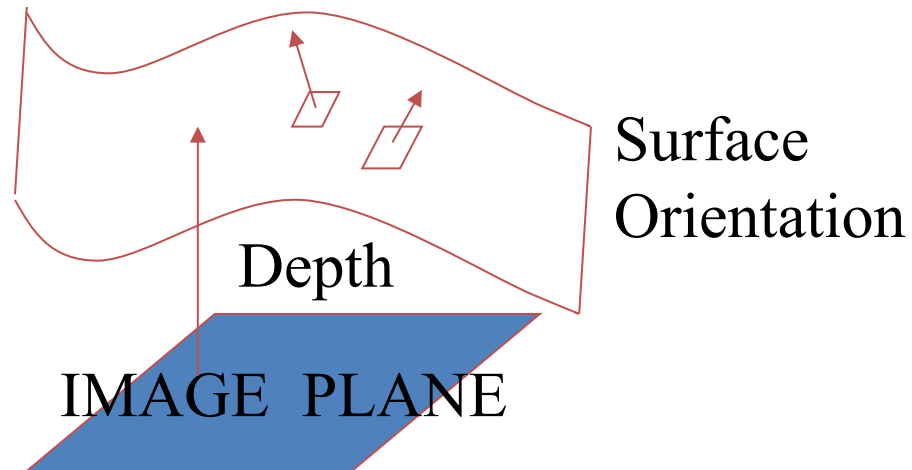
Normal Map vs. Depth Map

- Can determine Depth Map from Normal Map by integrating over gradients p, q across the image.
- Not all Normal Maps have a unique Depth Map. This happens when Depth Map produces different results depending upon image plane direction used to sum over gradients.
- Particularly a problem when there are errors in the Normal Map.



Normal Map vs. Depth Map

- A Normal Map that produces a unique Depth Map independent of image plane direction used to sum over gradients is called integrable.
- Integrability is enforced when the following condition holds:
$$\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}$$



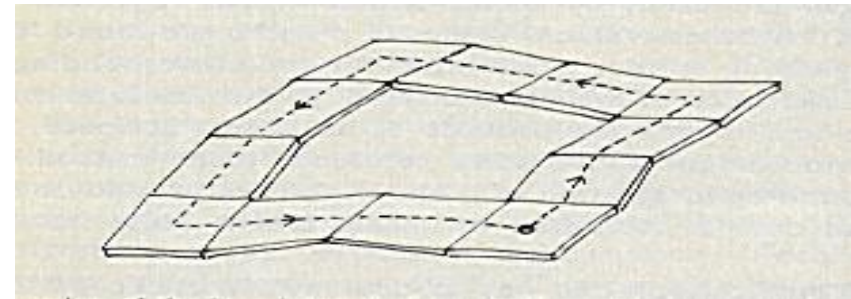
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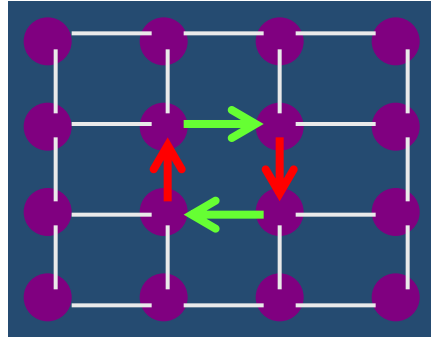
Green's Theorem

$$\iint (\partial p / \partial y - \partial q / \partial x) dx dy = \oint (p dx + q dy)$$



Integrability

- Gradient field of solution Z should be **integrable**
 - $\text{Curl}(Z_x, Z_y)$ (**loop integrals**) should be zero

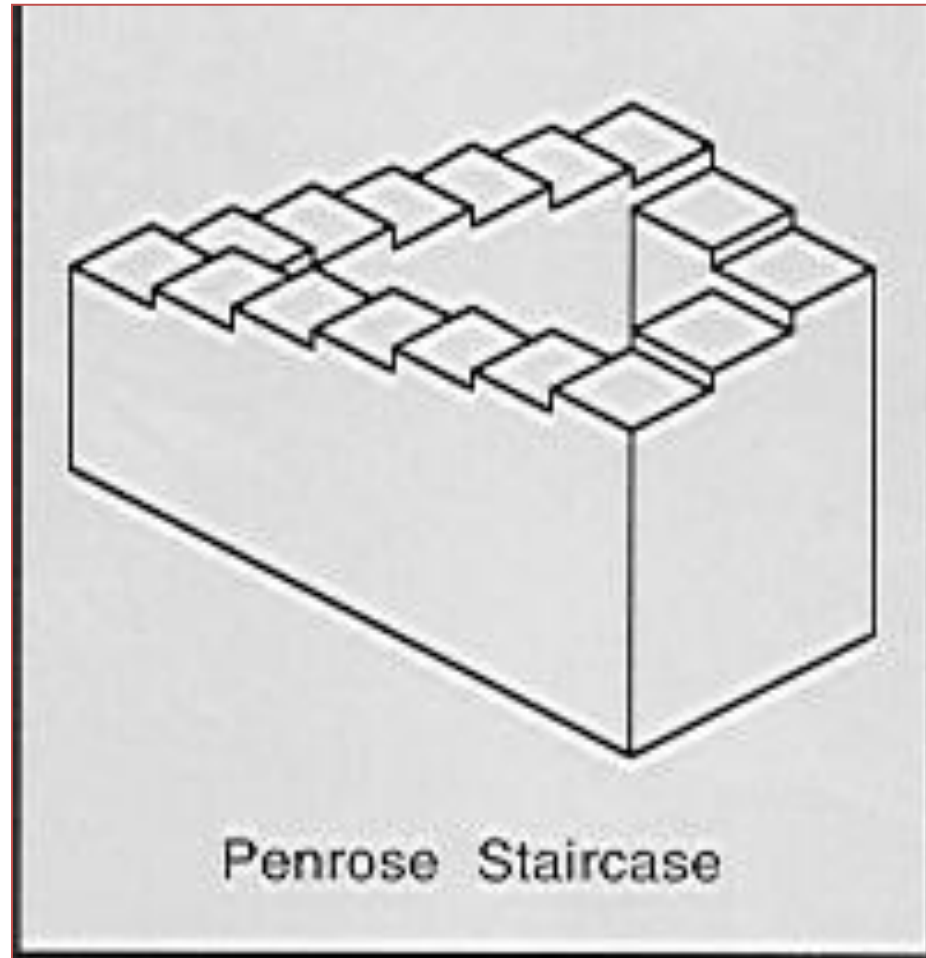


$$\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}$$

- Noisy estimated gradient field is **non-integrable**
 - $\text{Curl}(p, q) \neq 0$

Normal Map vs. Depth Map

Violation of Integrability

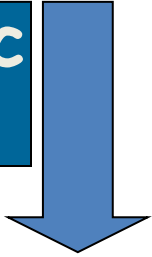


Height from Gradients: Photometric Stereo



Mozart Head images
under varying illumination

Photometric
Stereo

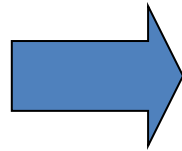


X Gradient

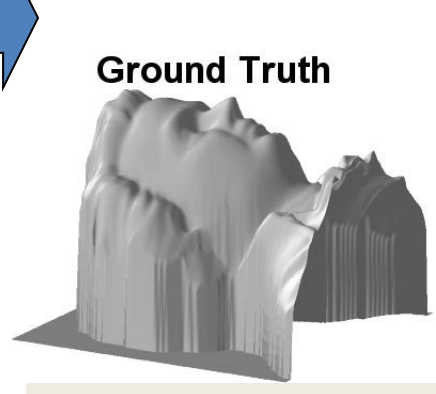


Y Gradient

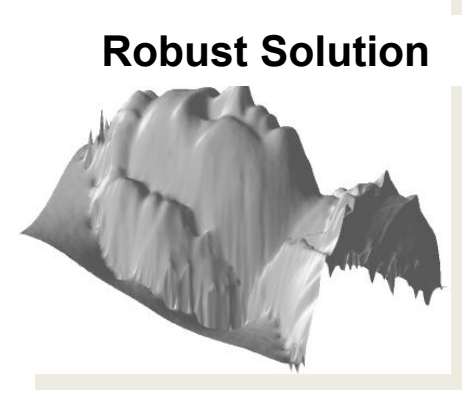
2D Integration



(Non-integrable Gradients)



MSE=2339.2



MSE=373.72

Practical Solution: Surface from Derivatives

- Goal: Compute the depth $z(x,y)$ of a surface from its measured derivatives $p(x,y)$ and $q(x,y)$.
- Estimate the derivatives of the surface $z(x,y)$ by

$$\begin{aligned}\hat{p}(x,y) &= (z(x+1,y) - z(x-1,y))/2 \\ \hat{q}(x,y) &= (z(x,y+1) - z(x,y-1))/2\end{aligned}$$

represented as matrix multiplication

- Find $z(x,y)$ that minimizes $E = \sum_{x,y} ((\hat{p}(x,y) - p(x,y))^2 + (\hat{q}(x,y) - q(x,y))^2)$

Using pseudo inverse ($1M$ unknowns for a $1K \times 1K$ image)

Practical Solution: Surface from Derivatives

Find the min-square solution to the set of $2M$ equations with $1M$ unknowns:

$$DZ = \begin{pmatrix} P \\ Q \end{pmatrix}$$

Where

- D : $2M \times 1M$ matrix representing the computation of p and q from z for each pixel
- Z : $1M$ vector representing the depth at each pixel
- P, Q : $1M$ vector each of measured derivatives

Limitations of Photometric Stereo

Big problems

- Doesn't work for shiny and colorful surfaces, semi-translucent surfaces
- Problems with shadows and inter-reflections (non convex objects)

Smaller problems

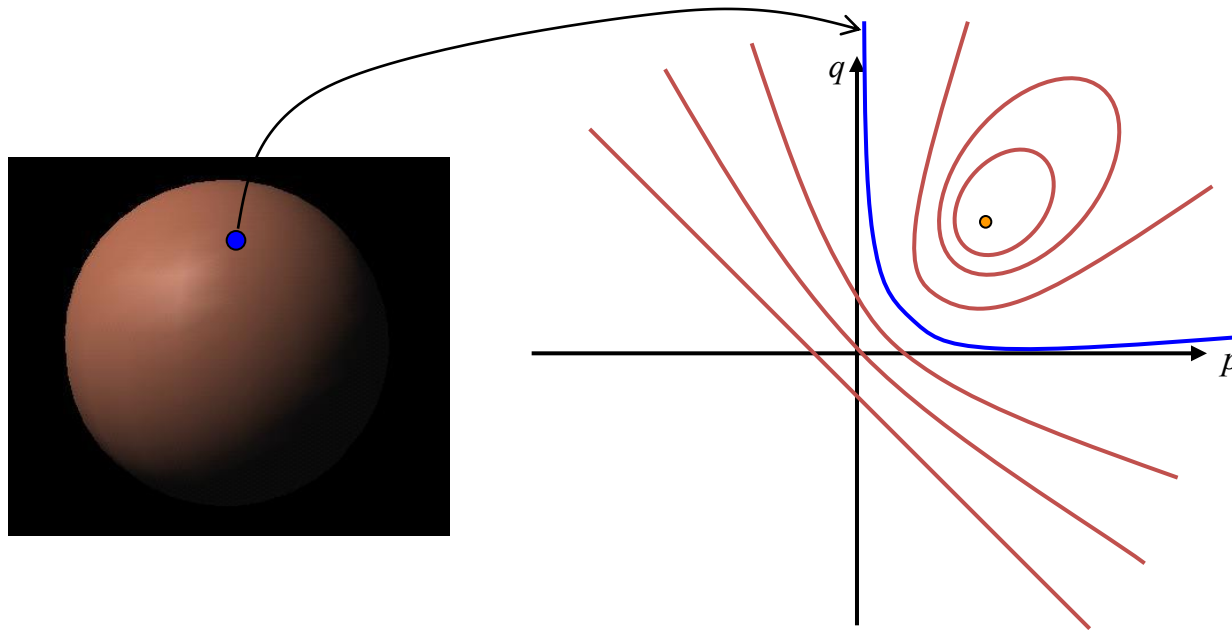
- Camera and lights have to be distant
- Calibration requirements
 - measure light source directions, intensities
 - camera response function

Shape-from-Shading

Single-Image Shape Recovery?

Given a single image of an object with known surface reflectance taken under a known light source, can we recover the shape of the object?

Given $R(p, q)$ (p_s, q_s) and surface reflectance) can we determine (p, q) uniquely for each image point?



NO

2 Approaches

Take more images with different light sources

- Photometric stereo

Add constraints

- Shape-from-shading

Image Irradiance Constraint

Find p and q for each image point such that the image irradiance should match the reflectance map

$$\iint_{object} (I(x, y) - R(p(x, y), q(x, y)))^2 dx dy$$

No Unique Solution!!!

Add Smoothness Constraints

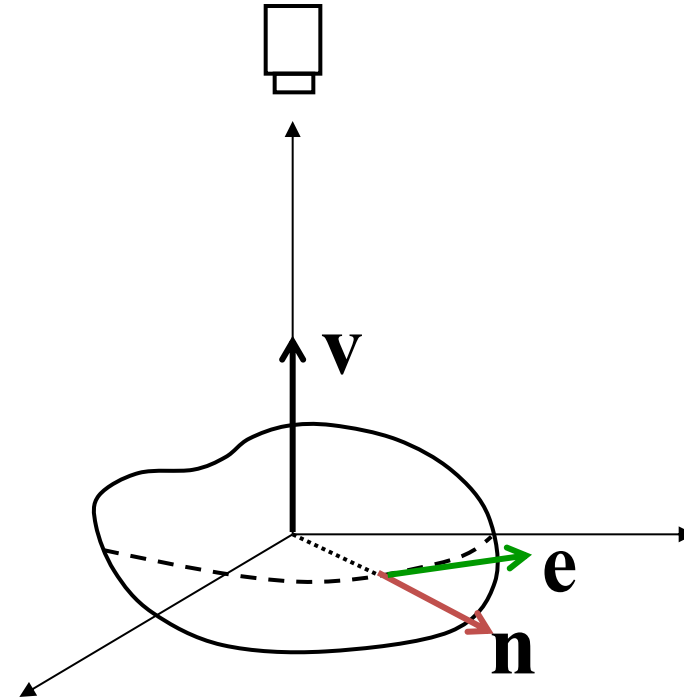
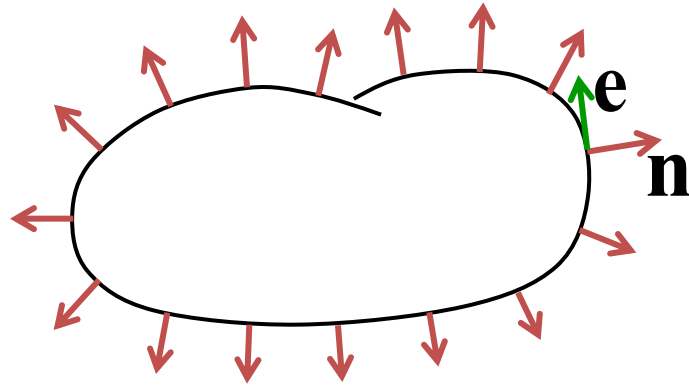
Assume that surface orientation (p, q) at neighboring points is similar.

Minimize:
$$\iint_{\text{object}} \underbrace{(p_x^2 + p_y^2 + q_x^2 + q_y^2)}_{\text{Smoothness Term}} + \lambda \underbrace{(I(x, y) - R(p, q))^2}_{\text{Data Term}} dx dy$$

Smoothness: minimize sum of square derivatives of p and q . Penalty for rapid change in p and q . Note: involves 1st and 2nd derivatives of z !

Additional Boundary conditions needed.

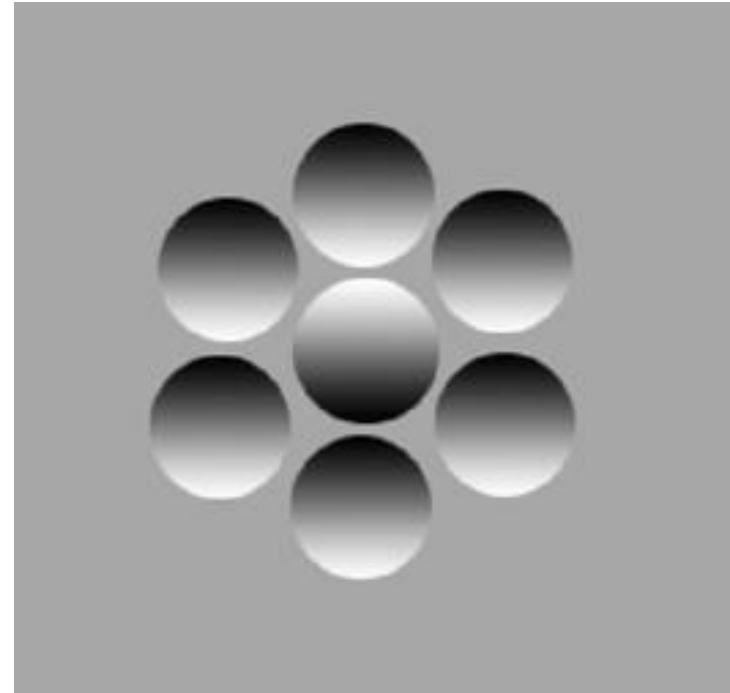
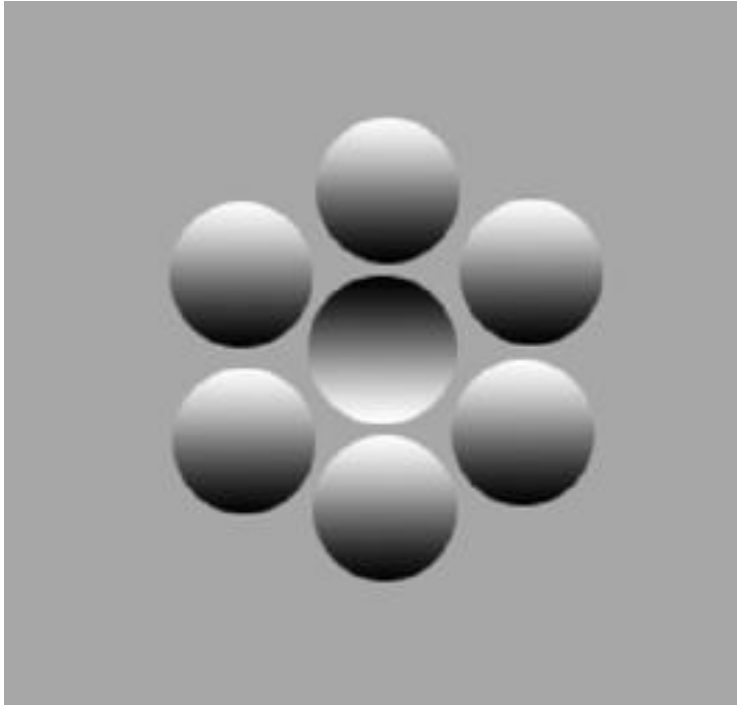
Boundary Conditions: Occluding Boundaries



$$\mathbf{n} \perp \mathbf{e}, \quad \mathbf{n} \perp \mathbf{v} \quad \therefore \quad \mathbf{n} = \mathbf{e} \times \mathbf{v} \quad \mathbf{e} \text{ and } \mathbf{v} \text{ are known}$$

The $\mathbf{n}$ values on the occluding boundary can be used as the boundary condition for shape-from-shading

Shape from Shading



- Intensity is a function of surface gradient & light-source location
- Interpretation depends on assumption for light-source location
 - E.g. We assume that lights comes from above...
 - Right image is same as Left just rotated 180°

Shape from Shading

